

Worth Fighting For: Daughters Improve their Mothers' Autonomy in South Asia

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Abstract

In South Asia, parents prize sons for both economic and cultural reasons, and having a son is often thought to improve his mother's status within the household. However, using data from Bangladesh and India, we show that such high regard does not necessarily translate into improved autonomy for her. In fact, a daughter raises her mother's participation in household decisions and her freedom of mobility relative to a son. A daughter also prompts her mother to work more, but not necessarily to consume more. These effects are strongest among mothers of older girls. These results are consistent with a theoretical model in which mothers have greater relative preferences for spending on their daughters than fathers do, and so seek more autonomy to direct resources to their daughters.

Keywords: Female autonomy, child gender, intra-household bargaining, female labor supply

JEL codes: J22, O12, J16, D13

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1 Introduction

Many mothers and fathers in South Asia report a preference for sons over daughters. Sons do not require dowries, are thought to be more likely to provide support to their aging parents, and are important in certain religious and cultural ceremonies. Accordingly, there is qualitative evidence that a woman's status within the household increases when she has a son. For instance, [Das Gupta et al. \(2003\)](#) argue that "a woman's main source of standing in her husband's family is as the mother of the future men of the family" (p. 172) and [Puri et al. \(2011\)](#) quote a woman who observes that "everyone in the family is very nice to a woman with a son because she has done her job" (p. 1171).

Despite this high regard, we document that in India and Bangladesh, a woman's participation in household decision-making and freedom of mobility (which we refer to as autonomy¹) actually increase after having a daughter, relative to a son. A woman's labor supply also increases, but her consumption (as proxied by her body mass index and the occurrence of anemia) does not. By contrast, her husband's labor supply falls after a daughter, relative to son. These patterns persist when we look at a sample of only first-born children – where we find no evidence of sex-selective abortion in either country – so our results do not seem to be driven by the fact that conservative households are less likely to have daughters.

These results are consistent with an intrahousehold bargaining model in which a mother cares more about her daughter's consumption than her husband does. We endogenize female autonomy and show that these differential preferences for spending on a daughter prompt a mother increase her autonomy (even when doing so is costly) and her labor supply in order to direct more resources to her daughter.² While daughters have many other effects on households in South Asia, we argue that models based on other effects of daughters – but without endogenous female autonomy – do not naturally explain the full set of empirical results. For instance, several models predict that daughters increase their mother's labor supply (daughters can assist home production, freeing their mothers to work outside the home, and daughters require dowry payments), but we argue that these models make no direct prediction on a mother's autonomy. Working itself could increase a mother's autonomy through improving her negotiation skills or her

¹In section 2.1 we discuss how these measures – and the way we model autonomy – relates to theoretical conceptions of autonomy in economics and other disciplines.

²By contrast, [Li and Wu \(2011\)](#) focus on women's participation in purchases of consumer durables for the household and find that sons increase their mother's participation in these decisions in China. Since these purchases are not necessarily targeted towards children – the most common include fans, television, and radio – we view this measure of bargaining power as capturing a different dimension of autonomy than what we model and examine empirically.

future earning ability ([Atkin 2009](#); [Anderson and Eswaran 2009](#)), but we present several pieces of suggestive evidence that labor supply is not the only channel linking daughters to increased autonomy, especially in India where we find that autonomy increases come before labor supply increases.

A well-known body of research has documented that resources allocated to mothers (or grandmothers) in developing countries differentially improve children's, especially girls', health and education ([Thomas 1990](#); [Duflo 2003](#); [Rangel 2006](#); [Qian 2008](#); [Doepke and Tertilt 2014](#)). We make a complementary point: women do not only dedicate a greater share of an exogenous income transfer to their daughters than men would. Instead, we argue that they also attempt to exert a greater control over household-wide division of resources if they have stronger preferences for spending on their daughters than their husbands do.

This is a different determinant of women's autonomy than has been highlighted in the economics literature. This literature has highlighted either the characteristics of a woman entering marriage, such as her age ([Jensen 2003](#); [Field and Ambrus 2005](#)), caste ([Field, Jayachandran and Pande, 2010](#)), or education ([Mocan and Cannonier, 2012](#)), or other determinants of her outside option such as her legal rights to property ([Field, 2003](#)) and inheritance ([Harari 2013](#); [Roy 2015](#)), job availability ([Majlesi, 2016](#)), or her own labor market experience ([Atkin 2009](#); [Anderson and Eswaran 2009](#)). We argue additionally that even holding constant a woman's characteristics and her institutional environment, she can choose to seek greater autonomy if her gains from doing so are sufficiently high. By extension, the fact that not all women choose to do shows that seeking additional autonomy is not without cost. Our model can then help explain the findings of recent randomized control trials that give women productive inputs or cash transfers but find little effect ([Blattman et al. 2016](#); [Haushofer and Shapiro 2013](#)) – or even reductions ([Roy et al., 2015](#)) – on standard measures of autonomy, such as their participation in household decision-making. We argue that even if the treatments increase a woman's outside option and could theoretically increase her autonomy, if exercising this autonomy is costly, then women will only do so if they perceive ways in which they would like to change household behavior.

Women's decision to seek autonomy – at a cost – has been described in qualitative research. [Agarwal \(1997\)](#) argues that social norms tend to dictate that a husband is the primary decision-maker, and apparent acquiescence on the part of women can reflect a strategic decision not to face the costs of upsetting these norms.³ However, she goes on

³This is in contrast to [Sen \(1990\)](#), who posits that women's preferences change as a result of their low bargaining power: "the underdog comes to accept the legitimacy of the unequal order and becomes an

to explain that if sufficiently motivated, women use tools such as withdrawing, playing relatives off one another, or withholding sex to influence household decisions. [Kabeer \(1999\)](#) describes this empowerment as “the process by which those who have been denied the ability to make strategic life choices acquire such an ability” (p. 435). We argue that a desire to direct household spending towards a daughter is one factor that motivates women to undertake this process. Indeed, like Agarwal, [Kabeer \(2000\)](#) describes women in Bangladesh who are in general happy with their husbands making decisions, but choose to exert control over finances when they refuse their responsibility. She cites a garment worker who explains that as “as long as men observe their responsibilities to their households, women consider the issue of who ‘controls’ their wages to be irrelevant.” (p. 149). Kabeer juxtaposes this account with profiles of several women who began work – and keep control of their income in “open defiance” of their husbands – because their husbands were neglecting their duties, specifically to fund their daughters’ welfare and education.

All together, this qualitative evidence suggests that the woman’s action to seek autonomy is important in explaining tendency for girls to increase autonomy and prompts our focus on the woman as the key actor in the theoretical model. At the same time, the idea of endogenous female autonomy can be viewed through a different lens, that the husband may grant autonomy when it is a daughter, if doing so saves him the cost of monitoring his wife’s spending. We characterize this model in section [2.3](#) and, while we provide some suggestive evidence that heterogeneity based on the mother’s baseline autonomy is more consistent with the baseline model, we acknowledge that realistically the increased autonomy seen among mothers of daughters is likely driven by both spouses’ choices.

The desire to spend on a daughter also raises a mother’s gains from working, and thereby increases her labor supply. [Heath and Tan \(2015\)](#) also argue that the gains from working increase female labor supply, but focus on inheritance rights that improve a woman’s bargaining power and thus affect her ability to control the income that she does earn. In this paper we provide a different – although complementary – explanation: the woman’s endogenous desire to control income (not just her exogenous ability to do so) also affects her labor supply.

Finally, our results also help explain the origins of mothers’ reported son preference, which in some cases is even stronger than fathers’ reported preference ([Jayachandran, 2014](#)). We argue that this preference does not necessarily reflect mothers’ deeply rooted animosity toward fellow females. Instead, in our model, mothers’ equilibrium utility is

implicit accomplice” (p. 126).

higher after a son than a daughter even if mothers get the same utility from the well-being of daughters and sons, as long as fathers prefer spending on sons. The mechanism is that mothers know that their husbands are unlikely to spend on daughters, so mothers have to decrease their own consumption and increase their autonomy (which is costly) in order to ensure that resources are devoted to daughters. Our findings then can shed some light on the relationship between autonomy, commonly measured as household decision-making, and women's fundamental well-being. If women are forced to take more responsibility – so they can direct proper resources to their daughters – the increased (costly) autonomy does not necessarily suggest an increase in women's well-being.

The rest of the paper proceeds as follows. Section 2 models the decisions of mothers and fathers to work and invest in children in a noncooperative household. In section 3 we describe the data and empirical strategy we use to test the predictions of the model. Section 4 provides results of these tests, and section 5 discusses the relationship of our results to models in which daughters affect their mother's labor supply directly. Section 6 concludes.

2 A Model of Endogenous Female Autonomy

2.1 Overview

In this section, we set up a simple model that identifies how a child's gender affects the mother's financial autonomy in a household setting. In accordance with recent literature that shows that bargaining outcomes between husbands and wives are not necessarily efficient (Duflo and Udry 2004; Robinson 2012; Heath and Tan 2015), we use a noncooperative model that does not impose Pareto efficiency, although in appendix B.5, we show that the main mechanism would be similar in a collective model (Browning and Chiappori, 1998). In this context, we assume that a woman starts with a baseline level of autonomy but can choose to seek more at a cost. That is, in order to get more say in how money is spent – including on children's human capital investments – the wife pays some direct utility cost. So the wife is willing to seek more autonomy only when her preferences are sufficiently different from her husband's.

There are several forms that the cost of seeking autonomy could take. It could reflect some sort of (non-monetary) compensation she gives her husband in exchange for greater control over resources. Alternatively, seeking autonomy could create conflicts, as suggested by papers that find that the economic empowerment of women in developing countries can lead to domestic violence (Eswaran and Malhotra 2011; Heath 2014; Ander-

son and Genicot 2015).⁴ Finally, it could represent a psychic cost of deviating from social norms that women do not control resources within the household (Bertrand, Kamenica and Pan, 2015).

It is worth discussing how autonomy (as we model and assess empirically) relates to theoretical conceptions of autonomy in economics and other disciplines. Anderson and Eswaran (2009) posit that “female autonomy is typically defined as the ability of women to make choices/decisions within the household relative to their husbands.” Self determination theory in psychology (Ryan and Deci 2000; Deci and Ryan 2012) defines autonomy similarly as an individual’s ability to make decisions that reflect his/her own values rather than those imposed by others, and argues that it is a key source of intrinsic motivation for actions. Of course, the ability to make decisions is fundamentally unobservable, so we follow Anderson and Eswaran (2009) in using measures of realized decision-making, such as whether a woman reports participating in decisions about household purchases. Even if these measures depart to some extent from the precise conceptual definition of autonomy, they still capture the key decision in our model, which relates to the wife’s decision to seek control over a certain share of financial resources within the household.⁵ Note also that our empirical measures of freedom of mobility (whether a woman needs permission from her husband to go various places) correspond more closely to the ability to make decisions about where to go rather than whether she chooses to make the decisions or not and thus closely reflect theoretical conceptualizations of autonomy.

We begin by laying out a baseline model of parental investments in their children and the wife’s decision to seek autonomy.⁶ In the baseline case, the only difference between sons and daughters is the utility their mother and father get from their well-being. Then in section B.1 of the appendix, we build in various other differences between sons and daughters. Specifically, we focus on four cases: daughters contribute time to help mothers with housework (extension I), daughters require dowry (extension II), daugh-

⁴Of course, women’s economic empowerment also improves their outside option, which allows women to either leave or credibly threaten to leave bad marriages and would (*ceteris paribus*) tend to decrease violence. Indeed, Heath (2014) finds suggestive evidence of these two countervailing effects. Our point is merely that if a woman seeks greater autonomy, violence could result, especially if the greater autonomy is not prompted by an increase in her outside option (as we assume in our theoretical model).

⁵While we could have used the term “control over household resources” throughout the paper to more precisely describe the behavior we capture, we call the key choice variable in the model “autonomy” given its correspondence to autonomy in the model of Anderson and Eswaran (2009). Other papers use the terms “bargaining power” (Atkin 2009; Mocan and Cannonier 2012), “decision-making power” (Majlesi 2016), “participation in household decisions” (Field 2003), or “status” (Jensen 2003), but all relate fundamentally to her participation in or ability to make decisions within the household.

⁶Because the model takes the presence of a child as given, we use woman/wife/mother and man/husband/father interchangeably. The variables related to the female always have subscript f and those related to the male always have subscript m .

ters lower the mother's cost of seeking autonomy (extension III), and daughters give a lower return to time and monetary investment (extension IV). In each extension, we show how this additional difference between sons and daughters affects the prediction on increased autonomy. Broadly, these extensions show that incorporating many other effects of daughters in our baseline model does not affect its key prediction that women seek higher autonomy after having daughters.⁷

One final interpretation consideration is whether the empirical relationship between daughters and their mothers' autonomy comes from the wife's decision to seek more autonomy with a daughter (as our model posits), or the husband's decision to grant her more autonomy. In section 2.3, we show that the husband's differential preference toward sons and daughters could indeed make him choose to monitor his wife closely if she has a son and grant her autonomy if she has a daughter. While we explain that the relationship between women's baseline autonomy and the effects of daughters is more in line with our main model – which also fits the qualitative evidence cited in section 1 – we acknowledge the net effect of daughters on autonomy is likely a function of the husband's actions as well.

2.2 Baseline Model

Suppose the wife's utility has the following form:

$$u_f(x_f, z, l_f, a_f) = \beta_f \ln x_f + \gamma_f \ln z + \delta_f \ln l_f + \alpha_f \ln \left(1 - \frac{a_f - \underline{a}}{\underline{a}}\right)$$

where her utility depends on her consumption x_f , the well-being of her child z , her leisure l_f , and a cost (α_f) to obtain a certain percentage increase in autonomy $\frac{a_f - \underline{a}}{\underline{a}}$.⁸ In particular, $\underline{a}$ is the wife's baseline autonomy that she gets for free, and a_f is the wife's actual autonomy such that the increase from the baseline $a_f - \underline{a}$ is costly. We choose to let the cost depend on the percentage increase in autonomy: that is, to obtain the same increase in autonomy, $a_f - \underline{a}$, it is less costly when the baseline autonomy is larger.⁹

Evidence for this assumption is given in table A1, which finds no evidence of het-

⁷One additional effect of daughters on autonomy could be that the high status afforded to mother of sons described in the introduction could directly translate into higher autonomy. If so, then this represents a countervailing force to our model's prediction that a mother seeks more autonomy with a daughter, and the increases we estimate represent the net effect of daughters inclusive of this direct effect of a son.

⁸We follow the setup in Anderson and Eswaran (2009) and examine a log linear (Cobb-Douglas) utility function to simplify the analysis. In appendix B.6, we confirm that the results persist with more general utility functions.

⁹Specifically, the marginal cost of additional autonomy, $-\frac{\partial u_f}{\partial a_f} = \frac{1}{2\underline{a} - a_f}$ is decreasing in $\underline{a}$.

erogeneous effects of a firstborn daughter on the mother's autonomy and labor supply based on the sex ratio of a state in India. Absent the tendency for women with higher baseline autonomy to have a lower cost of seeking additional autonomy, we would expect an unambiguously larger effect of daughters on autonomy in areas with sex ratios skewed towards males, where the difference in the husbands' preferences for sons over daughters is presumably larger and gains to seeking autonomy for women are accordingly higher. Moreover, in the sample of all births, there is a larger effect of daughters on autonomy in states with more women, providing further evidence that higher baseline autonomy makes it easier to fight for more autonomy. While this result should be interpreted with caution given the potential selection reflected in daughters beyond firstborn described in section 3.2, note that a selection mechanism that only already empowered women can have daughters is stronger in states with fewer women would work against finding this effect. Therefore, while we assume the particular functional form of marginal cost to fit the heterogeneity of results based on the sex ratio of a state, the key prediction that a daughter increases autonomy go through under a very general set of assumptions on the utility function, as we show in appendix B.6.

As will be seen in the budget constraints to each spouse's maximization problem (equations 1 and 2), the more autonomy the wife has, the more of her income she controls. The wife spends a unit of time on working outside the home e_f , working at home (taking care of the child) h_f , and leisure l_f , so $e_f + h_f + l_f = 1$. An alternative conceptualization of autonomy could be the wife's control of her time. In appendix B.2 we formalize this model and show that it also predicts that daughters prompt their mothers to seek more autonomy.¹⁰

The husband's utility has an analogous form:

$$u_m(x_m, z, l_m) = \beta_m \ln x_m + \gamma_m^{s,d} \ln z + \delta_m \ln l_m$$

where his utility depends on his consumption x_m , the well-being of his child z , and his leisure l_m . The husband spends a unit of time on working e_m and his leisure l_m , so $e_m + l_m = 1$.

In this baseline model, there is only one exogenous child in the household. Many of our key empirical results focus on the gender of the first-born child, which we show

¹⁰Our empirical measures of autonomy are a mixture of measures that relate more closely to financial control (a woman's participation in decisions about large household purchases) and those that relate more closely to control over time (whether she can visit her family alone). Others are a mixture of both (whether she can go to a health facility alone). In appendix table A2 we show the effects of daughters on each autonomy measure and we broadly find that daughters increase mothers' control over both financial and time resources.

in section 3.2 does not display any evidence of responding to selective abortion. When we turn to the gender of all children, our model suggests that the effects of subsequent daughters on autonomy could be lower, given that the marginal return on autonomy decreases once the mother's autonomy has already increased after the first daughter, and the marginal cost of autonomy is higher: $-\frac{\partial u_f}{\partial a_f} = \frac{1}{2a - a_f}$ is increasing in a_f .

To study the effect of child's gender, our baseline model assumes that the husband's utility from a son's well-being is higher than the utility from the daughter's well-being, $\gamma_m^s > \gamma_m^d$. Because the key to our model is the husband's relative preference for spending on a son relative to a daughter (compared to the wife's relative preference), we assume for simplicity that the wife's preference for child's well-being is independent of a child's gender.¹¹ Qualitative evidence from South Asia supports the assumption that women tend to have greater relative preference for spending on daughters than do their husbands (Kabeer 2001; Ahmad and Neman 2013), as does the quantitative finding elsewhere in the developing world that money in the hands of women differentially benefits girls' health (Thomas 1990; Duflo 2003; Qian 2008).

A possible extension to the model would allow the husband to pay a cost to reduce the wife's autonomy. We show in a related paper (Heath and Tan 2015) that giving women more autonomy can increase their labor supply and thus increase household income, which then improves the husband's utility under fairly general assumptions on the set up of a non-cooperative bargaining game. So it is not necessarily profitable for the husband to reduce the wife's autonomy, and we omit this option in our model. Even aside from the case where more autonomy for the woman benefits the husband, allowing the husband to respond to the wife's pursuit of greater autonomy would not change the main conclusions of the model unless the cost to the husband to do so is so small that he can completely counteract the wife's desire to seek more autonomy after having a daughter.¹²

The well-being of the child depends on the wife's labor contribution h_f and the financial contribution from both spouses $y_f + y_m$. In other words, the child can be viewed as a household public good produced by both spouses. We assume it follows a Cobb-Douglas production function:

$$z = f(h_f, y_f + y_m) = (h_f)^\lambda (y_f + y_m)^\rho$$

¹¹As we mention in the introduction, the assumption that the wife gets equal utility from the well-being of a son and a daughter is not inconsistent with the fact that they tend to report preferences for sons over daughters if these reported preferences represent the equilibrium utility the wife would have in each case and not parameters of the utility function.

¹²For instance, Eswaran and Malhotra (2011) argue that violence is costly to men, but they still sometimes use violence to thwart women's pursuit of higher autonomy. If we built in a similar mechanism, then unless the cost of violence is very small, the man would respond to – but not completely cancel out – the wife's pursuit of higher autonomy, and the key predictions of our model would go through.

in which $\lambda, \rho > 0$. If $\lambda + \rho = 1$, it has a constant return to scale.

The wife earns $w_f e_f$ from her labor force participation, but only controls a fraction of her income, determined by her autonomy a_f .¹³ The wife's optimization problem can then be summarized as follows:

$$\begin{aligned} \max_{x_f, h_f, y_f, e_f, a_f} \quad & \beta_f \ln x_f + \gamma_f (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_f \ln(1 - e_f - h_f) + \alpha_f \ln\left(1 - \frac{a_f - \underline{a}}{\underline{a}}\right) \\ \text{s.t.} \quad & e_f, h_f \in [0, 1], a_f \in [\underline{a}, 2\underline{a}], x_f, y_f \geq 0 \\ & p_f x_f + y_f \leq a_f w_f e_f \end{aligned} \quad (1)$$

where p_f is the price of the wife's private good and w_f is the wage the wife can earn. In this baseline case, we assume the only income source for both spouses is their earned income, which simplifies the problem by assuring interior solutions to labor supply for both spouses. In section 2.4 and particularly Proposition 2.4, we show that incorporating unearned income and thereby allowing for corner solutions in labor supply does not change the main conclusion of the model.¹⁴ This is important, since as table 1 shows, less than half of women work outside the home, and the main labor supply data we use provides data on labor supply on the extensive margin.

The husband's optimization problem is similar:

$$\begin{aligned} \max_{x_m, y_m, e_m} \quad & \beta_m \ln x_m + \gamma_m^{s,d} (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_m \ln(1 - e_m) \\ \text{s.t.} \quad & e_m \in [0, 1], x_m, y_m \geq 0, \\ & p_m x_m + y_m \leq w_m e_m + (1 - a_f)(w_f e_f) \end{aligned} \quad (2)$$

where p_m is the price of the husband's private good, w_m is the husband's salary, and the husband controls the rest of the wife's income, $(1 - a_f)w_f e_f$.

The key innovation of the model is letting the wife have the ability to fight for higher autonomy if it is profitable. This endogenous autonomy has a complicated bi-directional relationship with labor supply. On the one hand, a higher autonomy increases the gains from working and thereby incentivizes the wife to work more (Heath and Tan 2015). At the same time, a higher labor supply raises the gains from seeking autonomy. This

¹³We postulate the same set-up in Heath and Tan (2015). It can be viewed as the reduced form from a wide class of bargaining games. In Heath and Tan (2015), exogenous increases in inheritance rights increased a_f . Here we argue that if the wife wants to increase a_f , she can pay a cost to do so, but we do not take a stand on the exact structure of the bargaining process.

¹⁴We provide more discussions on unearned income in section B.3 in the appendix.

bidirectionality makes the model complicated and potentially non-monotonic. We thus introduce a critical value for the husband's preference on the child γ_m^* , which is useful when making monotone predictions.

An equilibrium is *interior* if all variables don't have corner solutions. We focus on interior solutions since when both spouse contribute money to the household public good (the well-being of the child), there is a natural link between their utilities and they bargain through it. The corner solution with only one spouse contributing money is discussed in appendix B.4.

Proposition 2.1. *Suppose the Nash equilibrium is interior. If the husband's preference for the public good is lower for a daughter than a son ($\gamma_m^s > \gamma_m^d$), then when the child is a daughter (relative to son):*

- *The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).*
- *The husband works less outside (e_m).*
- *The monetary investment to the household public good decreases ($y_f + y_m$), if $\gamma_m^d \leq \gamma_m^*$.*

All proofs are in appendix A. When the child is a daughter, the husband has a lower preference for spending on the child, prompting the wife to contribute more to the child's financial expenses. So she both works more and fights for a higher autonomy to control more of that income. The increase in her labor supply leads to a decrease in both time working at home and her leisure. The husband, on the other hand, can work less because he doesn't need to pay as much for the child's well-being.

There are two effects moving the monetary contribution to the household public good in opposite directions: a direct effect is the husband's decrease in his contribution, and an indirect effect is the wife's increase in her contribution after working more and fighting for a higher autonomy. As long as the husband's preference for the daughter is below the threshold $\gamma_m^d \leq \gamma_m^*$, the direct effect outweighs the indirect effect, since increasing the contribution is costly for the wife. In this case, the wife's private consumption and her overall utility are also lower (despite her higher autonomy), because she pays a cost for her autonomy and contributes more to the daughter's well-being.¹⁵

By contrast, if $\gamma_m^d > \gamma_m^*$, it is possible that the indirect effect outweighs the direct effect, and total financial contribution is actually higher with a daughter. This is due to

¹⁵See details in the proof of Proposition 2.1. The predictions on women's consumption and utility are in general ambiguous in the extensions we describe in section B.1. The woman's real income increases (due to both a higher autonomy and higher labor supply) would tend to increase consumption, but the financial contribution to the daughters lowers her consumption (and this autonomy has a utility cost to her).

the non-monotone behavior of e_f and a_f mentioned earlier, i.e., the reinforcement of labor supply and autonomy: higher autonomy increases the marginal utility from working and higher labor supply increases the marginal utility of autonomy. With a sufficiently strong reinforcement when $\gamma_m^d > \gamma_m^*$, the indirect effect could outweigh the direct effect, and total investment could actually be higher for daughters than sons. While we point out this theoretical possibility, we do not think it is particularly relevant empirically: while tests of whether spending on children differs by their gender in South Asia do not always detect differences (Duflo, 2005), other studies do find greater spending on boys (Kingdon 2005; Zimmermann 2012b), and we are unaware of any studies finding bias in expenditures towards girls. Thus our theoretical predictions consider the case where $\gamma_m^d \leq \gamma_m^*$, which both fits better with existing evidence on child gender and financial investments in South Asia and yields clearer predictions on autonomy and labor supply.

We also remark that daughters could prompt their mothers to take high-paying jobs. In particular, aside from choosing hours of work, suppose women can also choose between different types of jobs. Say there are high-paying jobs w_f^H as managers and supervisors, and low-paying jobs w_f^L as those in agriculture. Assume $w_f^H > w_f^L$, but there is a sunk cost (c) to get a high-paying job. The cost could be a search cost, a training cost and/or a cost to get the husband's consent. We argue that more women would want to work on high-paying jobs when the child is a daughter. This is because the wife seeks a higher autonomy when the child is a daughter by Proposition 2.1, which makes the high salary more attractive. So we get an additional prediction that daughters increase female labor supply particularly into high-paying jobs.

2.3 Alternative Framework: Husband Chooses the Level of Monitoring

In this section, we consider the husband's endogenous reaction to the sex of his child. That is, the husband may prefer to grant more autonomy to the wife, so he does not need to provide (costly) monitoring of the spending on the child when he does not care as much about the daughter.¹⁶

We modify the model to formalize this possibility: the husband can either directly invest in the household public good, y_m , which requires his monitoring with a cost α_m , or give a transfer t_m to the wife and let her decide the total monetary contribution to the household public good (the child's well-being). By choosing the latter case, he does not

¹⁶We thank the editor, Andrew Foster, for suggesting this alternative model.

need to pay a monitoring cost.¹⁷ The husband's optimization problem becomes:

$$\begin{aligned}
& \max_{x_m, y_m, t_m, e_m} \beta_m \ln x_m + \gamma_m^{s,d} (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_m \ln(1 - e_m) - \alpha_m 1_{\{y_m > 0\}} \\
& \text{s.t. } e_m \in [0, 1], x_m, y_m, t_m \geq 0, \min(y_m, t_m) = 0 \\
& p_m x_m + y_m + t_m \leq w_m e_m + (1 - \underline{a})(w_f e_f)
\end{aligned} \tag{3}$$

The wife's problem is similar to the baseline model except we remove her decision on much autonomy to seek in to focus on the effect of the husband's actions:

$$\begin{aligned}
& \max_{x_f, h_f, y_f, e_f} \beta_f \ln x_f + \gamma_f (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_f \ln(1 - e_f - h_f) \\
& \text{s.t. } e_f, h_f \in [0, 1], x_f, y_f \geq 0 \\
& p_f x_f + y_f \leq \underline{a} w_f e_f + t_m
\end{aligned} \tag{4}$$

Using our interpretation of female autonomy such that a_f measures the fraction of the wife's income that she can control, then a_f is higher when the husband is willing to give a larger transfer to his wife.

We first note that when the equilibrium is interior, the husband always prefers to use the transfer with no monitoring cost. Specifically, in any interior equilibrium, the household spends the same amount on the household public good regardless of whether the husband chooses to use $t_m > 0$ or $y_m > 0$, and thus he has no incentive to choose y_m , which costs α_m . To be consistent with the insight that the husband monitors for a son, i.e., $y_m > 0$, it implies that in such equilibrium $y_f = 0$ for a son. That is, the husband pays the monitoring cost, only because if he gives the transfer to the wife, she may not spend all of it on the son.

Proposition 2.2. *Suppose the Nash equilibrium is interior except $y_m > 0$ for a son implying $y_f = 0$. If the husband's preference for the public good is lower for a daughter than a son ($\gamma_m^s > \gamma_m^d$), then when the child is a daughter (relative to son):*

- The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).
- The husband works less outside (e_m).
- The monetary investment to the household public good decreases ($y_f + y_m$).

¹⁷This model is also closely related to extension III in appendix B.1, in which we model daughters lower the wife's cost of seeking autonomy, and in appendix B.2, in which we model autonomy as a control of time and the husband can grant free time to the wife so he does not pay a supervising cost.

The insight of Proposition 2.2 is very similar to that of Proposition 2.1: Since the husband does not value the daughter as much as the son, he works less and contributes less to the child. Thus, the wife works more outside to invest in the daughter and less at home. She gets more autonomy because the husband does not want to monitor and grant more control of money to her.

We make two remarks here. First, the true model could be hybrid model in which the wife can seek autonomy and the husband can grant it. In any interior equilibrium of this hybrid model, both spouses contribute to the increased female autonomy. Moreover, the wife gets a higher utility in this hybrid model than that in the baseline model, because part of the autonomy is granted and she does not need to pay the entire cost of the increased autonomy.

Secondly, while these two models yield very similar overall predictions on women's autonomy and labor supply, they given different predictions on potential heterogeneous effects based on the mother's baseline autonomy. In our baseline model, the increase in autonomy is higher when the baseline autonomy is higher, because the cost of a given increase in autonomy, $a_f - \underline{a}$, is decreasing in $\underline{a}$. The model in which the husband monitors delivers the opposite prediction. When the baseline autonomy is higher, the wife has control over more income and she is more likely to contribute to the child's well-being: $y_f > 0$. Then, the husband prefers to give her a transfer knowing she will spend all of it on the child. So when the baseline autonomy is higher, the husband grants autonomy regardless of the child's gender, and daughters are less likely to increase autonomy. Thus, while we think both channels are likely present, the fact that in table A1 we see either a net zero effect or if anything a stronger effect of daughters on autonomy in states with sex ratios favoring women (plausibly proxying for a higher $\underline{a}$) provides some suggestive evidence that the mechanism in our main model is empirically relevant.

2.4 Focusing on Women Who Do Not Work Outside the Home

In our baseline model, a woman seeks higher autonomy for two reasons: one is driven by preferences and the consequent desire to spend on things she values more than her husband (such as a daughter), the other is that any increase in her income raises her marginal returns to autonomy. While the latter channel relates daughters to mother's autonomy in an endogenous autonomy model, we provide suggestive evidence that differential preferences for spending on daughters are at least part of the mother's decision to seek autonomy after having a daughter. To do so, we consider a restricted scenario in which women do not have the option to work outside the home. If women still seek

higher autonomy after having a daughter in this scenario, this autonomy would be used to change the spending of a fixed budget rather than assert control over income from increased female labor supply prompted by the daughter. Focusing on women who cannot work also separates the endogenous decision to seek more autonomy from the possibility that working more (due to other effects of a daughter) increases baseline autonomy (Atkin 2009; Anderson and Eswaran 2009).

Of course, the women we observe empirically who do not work have endogenously chosen to do so. If the decision to work is related to their baseline autonomy (as in Heath and Tan 2015), focusing on this does not give the causal relationship between daughters and mothers' autonomy if women did not have the option to work (as we will model theoretically). So we first use our model to assess how this selection effect would affect an empirical assessment of the relationship between daughters and mothers who do not work. In order to do so, we add a new components to the model: the wife's unearned income R_f .¹⁸ The wife's maximization problem becomes:

$$\begin{aligned} \max_{x_f, e_f, h_f, y_f, a_f} \quad & \beta_f \ln x_f + \gamma_f (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_f \ln(1 - e_f - h_f) + \alpha_f \ln(1 - \frac{a_f - \underline{a}}{\underline{a}}) \\ \text{s.t. } \quad & e_f, h_f \in [0, 1], a_f \in [\underline{a}, 2\underline{a}], x_f, y_f \geq 0, p_f x_f + y_f \leq a_f(w_f e_f + R_f) \end{aligned}$$

Proposition 2.3. *Suppose the Nash equilibrium is interior. If the baseline autonomy increases, the wife works more outside the home (e_f).*

As the wife gets a higher baseline autonomy, she gets a higher control over her earnings so it is optimal for her to work more. Suppose there are two thresholds t^d and t^s : If the wife's baseline autonomy is higher than t^d (or t^s) she goes out to work when it is a daughter (or a son). Since we predict a daughter increases the wife's incentive to work, $t^d < t^s$. When we focus on women who don't work, we compare women with $\underline{a} < t^d$ with a daughter to women with $\underline{a} < t^s$ with a son. If anything, this selection bias itself makes women with a daughter show a lower autonomy. With this potential bias in mind, we focus on women who don't work and re-examine the possible effects of a daughter.

Proposition 2.4. *Suppose the Nash equilibrium is interior and the wife gets an unearned income R_f but cannot work outside the home. We examine the effect of a girl, relative to a son, on the wife's autonomy (a_f), her labor supply at home (h_f), the husband's labor supply (e_m), and monetary investment to the public good ($y_f + y_m$):*

¹⁸Building in unearned income is necessary when we focus on the extensive margin of labor supply because otherwise she would have zero consumption – and infinitely negative utility – when she does not work.

- If the husband has a lower preference for spending on a girl ($\gamma_m^s > \gamma_m^d$), a daughter leads to higher a_f , the same h_f , less e_m , and less $y_f + y_m$.

In both the baseline model and alternative framework where the husband has a lower preference for spending on a daughter, it prompts the wife to seek more autonomy – even when she cannot work outside the home – in order to counter the decreased spending from her husband on the daughter. If she can work outside the home, the autonomy is also driven by the desire to increase control over increased income prompted by her daughter.

2.5 Summary of Testable Predictions of Theoretical Model

The key testable predictions on the wife’s behavior in the baseline model (and magnified by the extensions in which daughters help with housework or require dowry) are increased autonomy and labor supply, as the mother seeks to provide more resources to her daughter. The net effect on her own consumption is ambiguous: she brings in more income and controls a greater fraction of the income that she does bring in, but she also contributes more to her daughter given her husband’s lower contribution.

The model also has predictions on the behavior of the husband and resulting contribution to the household public good. In particular, male labor supply falls as the father invests less in the daughter. As a result, the baseline model predicts that the monetary investment in girls will generally be lower than investment in boys, since the mother’s increased investment is unlikely to compensate for the father’s decreased investment. Time investment is also lower in girls as mothers devote more time to working outside the home. This prediction also holds in unitary household models if girls represent a negative income effect.

3 Empirical Strategy

3.1 Data and Measures of Autonomy

For India our main measures of women’s autonomy come from the National Family Health Survey (NFHS) of 2005-2006. For Bangladesh, we use the 2007 and 2011 rounds of the Demographic and Health Survey (DHS). Each survey includes a nationally representative sample of ever-married women between the ages of 15 and 49. Thus, we cannot estimate the effects of child gender on older women, or on the small sample of births that

are to women who have never been married.¹⁹

Because the National Family Health Survey is designed to be comparable to DHS surveys conducted worldwide, several of the questions we use to capture autonomy are asked in both surveys. In particular, both surveys ask women who typically makes decisions about the woman's own health care, large household decisions, and purchases for daily life; possible options are the husband, wife, or both. The India survey additionally asks who makes decisions about spending the husband's earnings and visiting relatives. It also contains measures of a woman's freedom of mobility, in particular, whether she can go alone to the market, health facility, or places outside her village. Finally, it asks whether she has access to a bank account.

Table 1 provides summary statistics for sampled women with at least one child in both countries. In both countries this sample of women tends to have low education (5.1 years in India and 4.7 years in Bangladesh) and were married at a young age (17.8 years in India and 15.4 in Bangladesh). Women in India are more likely to work outside the home than women in Bangladesh (35 percent of women with at least one child in India, versus 19 percent of women in Bangladesh). Nonetheless, in the decision-making variables that were asked in both countries, women describe similar levels of autonomy; between sixty and seventy percent of women in both countries report some say in decisions about their own health care, large household purchases, and daily household purchases.

Because we have many outcomes that reflect autonomy, our main outcome of interest is an index of the individual autonomy measures. For the decision-making variables, we focus on whether a woman has some power over each decision.²⁰ We construct this index by normalizing each variable so that it has mean zero and standard deviation one, and then summing the normalized variables. In appendix table A2 we provide the estimation results of child gender on each component of the index to show that the results are not just driven by a small subset of the autonomy measures.

3.2 Estimating Equations and Identification Concerns

We estimate the effects of a child's gender on her mother's autonomy and labor supply (on the extensive margin) using two different specifications. The first uses the gender of the entire sample of a mother's children. So for a woman i , we test whether the number

¹⁹Fledderjohann et al. (2014) report that only 2 percent of births in India were to unmarried mothers.

²⁰The fact that a woman has sole control over a decision (relative to having some say in) may not capture additional bargaining power if this outcome reflects the fact that her husband leaves her on her own to control the household.

	India	Bangladesh
age	32.879	31.51
years of education	5.139	4.72
age at first marriage	17.829	15.47
children	3.04	2.95
bmi	21.495	21.36
works outside home	0.351	0.191
say in own health care	0.683	0.628
say in large household purchases	0.606	0.601
say in visits to family	0.674	
say in purchases for daily needs	0.684	0.632
say in spending husband's earnings	0.732	
can go to market alone	0.635	
can go to health facility alone	0.592	
can leave village alone	0.468	
has access to a bank account	0.188	
N	79,827	25,036

Table 1: Summary Statistics

of daughters affects an outcome Y_i conditional on her total number of children:

$$Y_i = \beta_1 Children_i + \beta_2 Daughters_i + x_i' \delta + \epsilon_i \quad (5)$$

The mother-level control variables in $x_i' \delta$ are a set of variables that are very unlikely to change in response to overall fertility or the children's gender: the mother's years of education, state of residence (India)/region of residence (Bangladesh), whether she lives in a rural area, whether she is Hindu, and dummies for her year of birth and whether she is a member of a scheduled caste, scheduled tribe, or other backward caste (in India).²¹ Our use of the continuous number of daughters as the primary independent variable of interest corresponds to the assumption of a constant marginal effect of daughters on autonomy. While we explicitly test this assumption in section 4.1 and find some evidence of nonlinearities, we nevertheless think that the average effect of a daughter is a useful benchmark.

Importantly, $Children_i$ and $Daughters_i$ report total children ever born to the women,

²¹Results are almost identical if we also include controls for her husband's age and education.

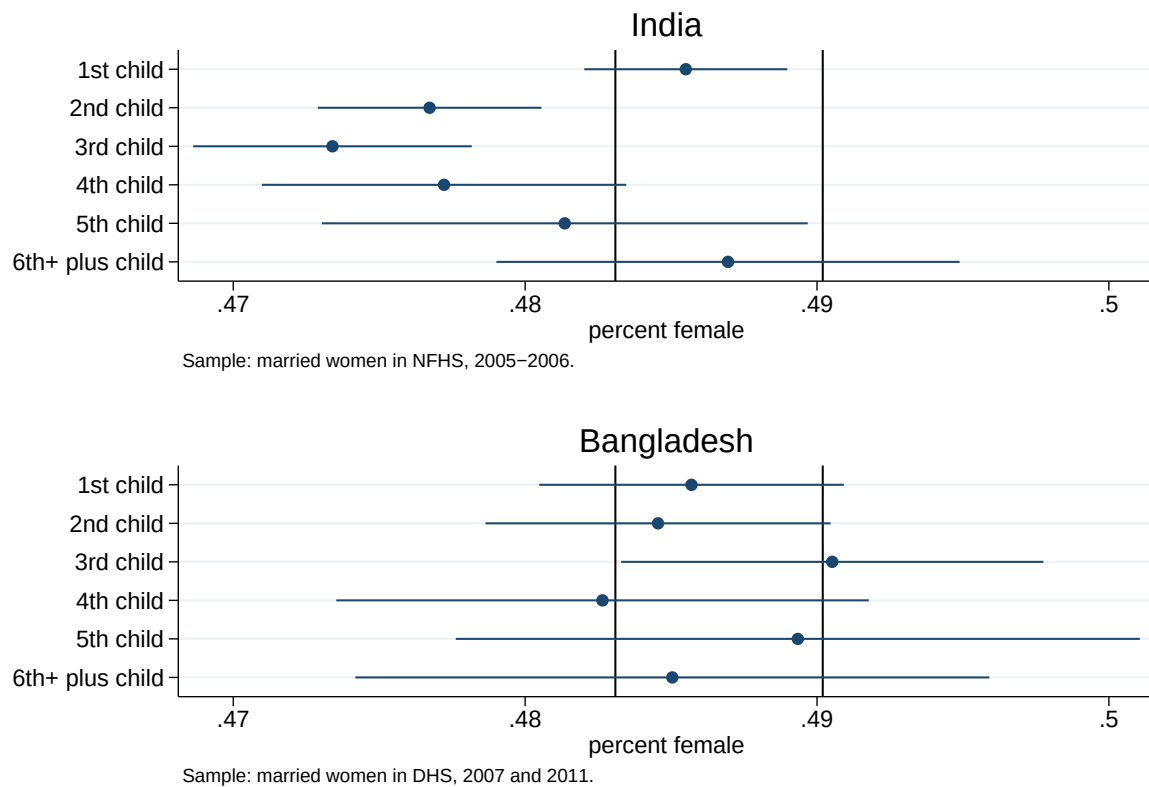
regardless of whether the children are still alive at the time of the survey, which avoids the reverse causality concern that would arise if only mothers who have high autonomy already can dedicate enough resources to daughters to ensure that they survive. This equation then estimates the causal effect of child gender on the mother's outcomes under the assumption that conditional on the number of children and their order, the gender of a child at birth is random. While the spread of ultrasound technology that facilitates sex-selective abortion may cause this assumption to fail – a possibility we discuss in detail below – we nonetheless think that a useful point of departure is to use all of the variation in gender that we observe.

To address the possibility that unobserved household-level factors drive both women's outcomes and the sex ratio at birth of their children in several ways, we consider the sample of first births, where previous research has found little evidence of sex selective abortion (Pörtner 2015; Bhalotra and Cochrane 2010; Rosenblum 2013). Indeed, figure 1 shows that 48.5 percent of first births in India are female, which is within the natural range (Chahnazarian, 1988). In Bangladesh, the confidence interval for all births lies within the natural range. The fact that the reported sex ratio is with the natural range also alleviates concerns about selective recall of children – particularly daughters –who are no longer living.

By contrast, subsequent births in India do fall outside of this range, suggesting sex-selective abortion or differential recall of children who are no longer living, given that excess female mortality is concentrated at higher parities (Gupta, 1987). To help understand the effects that this potential selection may have in equation 5, we first interpret in any potential selection in light of our model, which predicts that women with higher baseline autonomy (a) have less disutility from a daughter, since they can devote more resources to them without needing to pay the cost of higher autonomy. Thus, accordingly to our model, there could be upwards bias in the effect of daughters given by β_2 in equation 5.

Figure 2 assesses this prediction empirically by examining the relationship between parental and household characteristics and the probability that a child is a girl. In India, the largest determinant is wealth: a one standard-deviation increase in wealth decreases the probability that a child is a girl by 0.73 percentage points. This pattern likely reflects both their better access to ultrasounds and the fact that their smaller desired family increases son bias (Bhalotra and Cochrane, 2010), and also the possibility that biological channels such as poor nutrition during pregnancy (Almond and Mazumder, 2011) may cause boys to be missing in poor families.²² Regardless of the cause, because wealth is

²²Moreover, since wealth is measured at the time of the survey and not at the time of the child's birth,



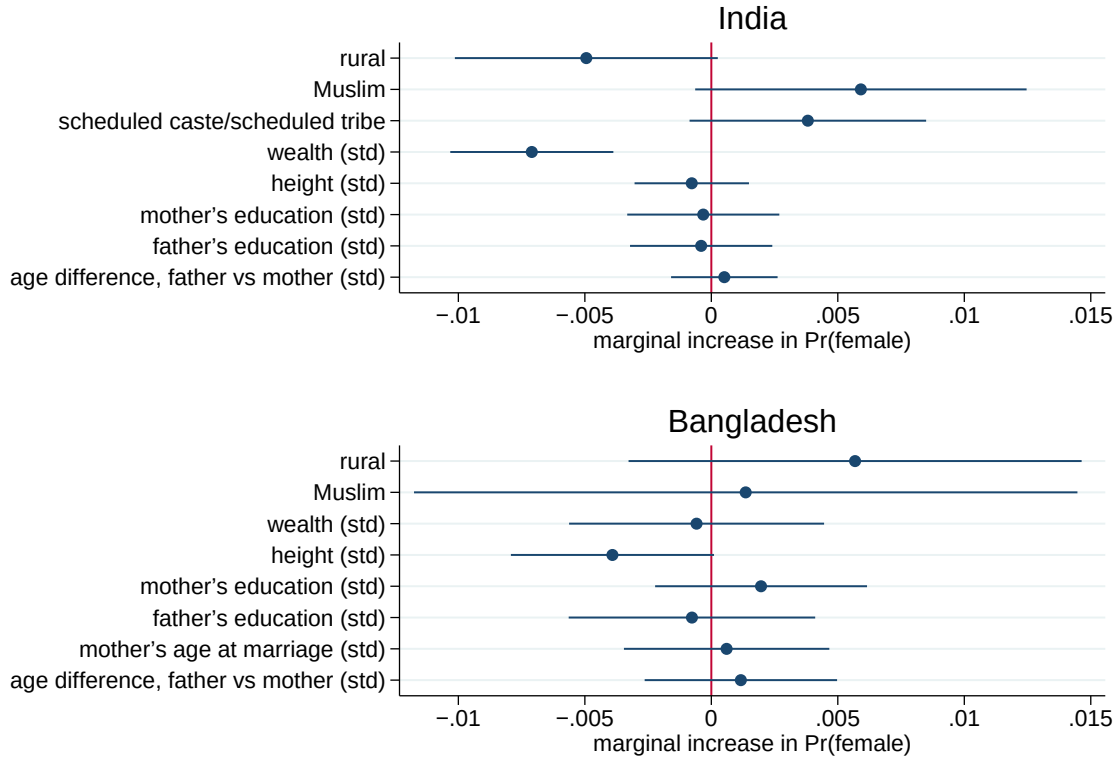
Vertical lines denote natural range of sex ratio at birth in Chahnazarian (1988)

Figure 1: Sex ratio, by birth order

associated with higher female autonomy – a one standard deviation increase in wealth corresponds to a 0.26 standard deviation increase in autonomy ($P < 0.001$) – it is likely that any bias from wealth or related measures of socioeconomic status would, *ceteris paribus*, bias downwards β_2 in equation 5. Therefore, the net bias on β_2 – this downward bias, versus upward bias from the relationship between baseline autonomy and the mother’s cost of a daughter pointed out by the model – is ambiguous. Urban households,²³ Muslim households, and households from scheduled castes and scheduled tribes are marginally more likely to have female children. Interestingly, conditional on the other controls, neither mother’s nor father’s education increases the probability of a girl.

The evidence of sex-selective abortion in India for births after the first and the am- these channels appear to dominate the fact that daughters increase savings as households save for their dowries (Deolalikar and Rose, 1998).

²³When we eliminate the other control variables, the sign of the coefficient reverses: rural households are more likely to have daughters, driven largely by the fact that they are less wealthy and disproportionately lower caste.



(std) denotes variables whose units are expressed in standard deviations

Figure 2: Correlates of a female birth

biguous nature of the bias in the estimated β_2 in equation 5 together motivate our second empirical strategy, which involves re-estimating equation 5 using only the gender of the firstborn child:

$$Y_i = \beta FirstBirthFemale_i + x_i' \delta + \epsilon_i \quad (6)$$

This gives the causal effect of having a first-born girl on the mother's outcome, under the assumption (supported by figure 1) that there is no selection in the gender of the first child.

Even if child sex at birth is random, there is considerable theoretical justification and empirical evidence that South Asian families change subsequent behavior in response to the sex of a child. For instance, son-biased stopping rules for fertility predict that parents are more likely to have another child after the birth of a girl than a boy (Clark 2000; Jensen 2003; Rosenblum 2013). The prevalence of dowries and lower anticipated lifetime transfers also dictate that the birth of a daughter (relative to son) decreases a family's permanent income (Deolalikar and Rose, 1998). Thus, if we continue to include only

predetermined controls in $x'_i\delta$, the effect of child gender on household outcomes reflects both direct responses to the child's gender and indirect effects of actions a household taken by a household to reoptimize after learning a child's gender.

Some authors have argued that examining outcomes in a short time horizon after a child's birth limits the opportunity that households have to reoptimize in response to a child's gender and therefore isolates direct behavioral responses to the gender of the child (Zimmermann 2012a; Barcellos, Carvalho and Lleras-Muney 2014). However, because the differential preferences in the main model may be especially salient at different ages of the daughter, we consider the overall effect of child gender at all ages of the offspring, inclusive of these indirect effects. Therefore, we do not limit our sample to the mothers of very young children, but instead, in appendix section B.1 we built these behavioral responses into our model and show that in general the main prediction that mothers seek more autonomy after having daughters remains. Relatedly, in section 5 we will argue that behavioral responses to the birth of a daughter are unlikely to explain the entire increase in mother's autonomy unless we also consider the endogenous autonomy response we model. Note, however, that because equation 5 estimates the effect of the number of daughters conditional on total fertility, this equation shuts down the effect of daughters on mothers' subsequent fertility as a potential channel linking daughters to autonomy.

The ages of children indeed matter for the relationship between child gender and mother's autonomy. Using the same India data as we do but focusing on only the mothers of young children, Zimmermann (2012a) finds small positive effects of a having a boy on a mother's autonomy in the child's first months of life, but these effects disappear once the child is 6 months old. Our overall finding that daughters increase their mother's autonomy suggests that the autonomy sought by mothers of older daughters eventually dominate the short-term boost in status a mother enjoys after having a boy. In section 4 we explicitly consider how the effects of child gender vary by the child's age and confirm that the positive effect of daughters on mother's autonomy is driven by older daughters.

4 Results

4.1 Key Mechanism: Autonomy and Female Labor Supply

In table 2 we present results from equations 5 and 6 that estimate the effects of a daughter on her mother's autonomy and labor supply in India and Bangladesh, first using all births and then just the first birth. In both countries and in both samples, having a daughter

leads to positive and statistically significant increases in her mother’s autonomy.²⁴ To contextualize the size of the effect, we compare it to the coefficient on an additional year of education. In India, the autonomy index increases by 0.058 units for each child that is a daughter, relative to a son, (versus 0.156 for an additional year of education), and if her first child is a girl a mother’s autonomy index increases by 0.115 units compared to when her first child is a son. Note that the sum of the positive coefficient on the number of daughters and the negative coefficient on children overall is slightly positive (although not significantly different from zero), implying that the negative correlation between the number of children she has and a mother’s autonomy is driven entirely by sons.

The larger coefficient on a daughter in the sample of first births is consistent with the model’s prediction that the effect of a firstborn daughter on autonomy is higher, given potential decreasing marginal returns to additional autonomy or increasing marginal costs, and confirmed in table A5 when we explicitly allow the effect of daughters to vary by sibling order and composition. Of course, given evidence of sex selective abortion in births after the first discussed in section 3.2, another potential difference between the effects of firstborn and later daughters is selection bias. If selection bias is a factor causing the larger coefficient on later daughters, this suggests that the tendency of women with higher baseline autonomy to avoid sex selective abortion – as predicted by the model – dominates the fact that women in wealthy families with access to ultrasound tend to have higher autonomy as reflected in figure 2.

In Bangladesh, the effect of a daughter, relative to a son, on the autonomy index is actually several times larger than the increase from an additional year of education: 0.083 for each child that is a daughter (versus 0.022 for each additional year of education), and 0.101 in the sample of the first child. While, unlike in India, the overall effect of a daughter on autonomy is negative ($P = 0.093$), it is still several orders of magnitude lower for a girl (-0.028 units) than for a boy (-0.111 units).

Turning to labor supply, in India each daughter is associated with a 0.46 percentage point increase in the probability that her mother works outside the home (relative to the effect of a son); a firstborn daughter is associated with a 0.84 percentage point increase. In Bangladesh the effects are again even larger. Each daughter there is associated with a 1.20 percentage point increase in the probability that her mother works, while a firstborn daughter increases the probability that her mother works by 1.64 percentage points.

In table A3 we break down the labor supply by the type of job that a woman enters. A consistent pattern emerges: daughters prompt women to work in jobs outside of the home

²⁴Standard errors are clustered at the level of the primary sampling unit. Heteroskedasticity-robust (Huber-White) standard errors give almost identical results.

	India		Bangladesh	
	Autonomy index	1(Work)	Autonomy index	1(Work)
	0.0584** [0.0268] -0.0312 [0.0262]	0.0046* [0.0025] 0.0000 [0.0023]	0.0829*** [0.0214] -0.1108*** [0.0198]	0.0120*** [0.0034] -0.0229*** [0.0030]
daughters				
children				
first birth female	0.1153** [0.0455]	0.0084** [0.0042]	0.1010*** [0.0352]	0.0164*** [0.0058]
mother's education	0.1556*** [0.0076]	-0.0124*** [0.0007]	0.0218*** [0.0042]	-0.0039*** [0.0009]
Mean Dep. Var.	0.000	0.351	0.000	0.192
Std Dev of Dep. Var.	5.233	0.477	2.500	0.393
Observations	79,827	79,686	25,028	25,025
R-squared	0.198	0.145	0.052	0.102
		0.145	0.048	0.037

Notes: Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include age fixed effects, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh). The autonomy index in India is an aggregate of whether the woman has say in large household decisions, purchases for daily life, her own health care, visits to family, and spending her husband's income. Additionally, it includes whether she has access to a bank account and whether she can go alone to the market, the health clinic, and locations out of town. In Bangladesh the autonomy index is an aggregate of whether the woman has say in large household decisions, purchases for daily life, and her own health care. (See section 3.1 for details).

Table 2: Effects of Child's Gender on Mother's Autonomy and Labor Supply

throughout the year, for cash, and for another person (rather than for themselves or for a family member). Furthermore, these jobs are in professional, clerical, or manual sectors, but not agriculture.²⁵ In [Heath and Tan \(2015\)](#) we find that an inheritance law reform increased women's autonomy, which increased their labor supply into these jobs, which we argued were likely to be high-paying. Increased labor supply into high-paying jobs is consistent with the remark of our baseline model, in which there is a cost to obtaining higher wage jobs, which women are more likely to seek out after having a daughter.

We now assess whether the effects of child gender vary with the age of the child. Specifically, we allow the effects of child gender in equations 5 and 6 to vary based on five-year bins, from birth to age 25:

$$Y_i = \sum_{a \in \{0,5,\dots,20\}} \beta_{1,a} 1(\text{Mother has child aged } a \text{ to } a + 4) \quad (7)$$

$$+ \sum_{a \in \{0,5,\dots,20\}} \beta_{2,a} 1(\text{Mother has daughter aged } a \text{ to } a + 4) + x_i' \delta + \epsilon_i$$

$$Y_i = \sum_{a \in \{0,5,\dots,20\}} \beta_{1,a} 1(\text{Firstborn child aged } a \text{ to } a + 4) \quad (8)$$

$$+ \sum_{a \in \{0,5,\dots,20\}} \beta_{2,a} 1(\text{Firstborn daughter aged } a \text{ to } a + 4) + x_i' \delta + \epsilon_i$$

Figures 3 and 4 graph the $\beta_{2,a}$ coefficients for autonomy and labor supply, for India and Bangladesh, respectively. Figure 3 shows that the effects of a daughter in India are driven by older girls, roughly beginning at age 10 to 14 (although this pattern is less strong in the sample of first children) for autonomy and ages 15 to 19 for labor supply.²⁶ In Bangladesh, shown in figure 4, a broadly similar pattern emerges, although the results are more consistent for autonomy than for labor supply, especially in the sample of just the oldest children. Note that because we continue to include year-of-birth dummies for the mother, the identification comes from the gender difference of children of different ages and not from age-patterns in autonomy.

In the context of the model, the fact that autonomy increases do not come immediately after birth could reflect the fact that parents' differential preferences for spending on girls versus boys are stronger for older girls than younger girls. For instance, parents

²⁵Our result stands in contrast to [Rose \(2000\)](#), who found that daughters increase their mother's labor supply in agriculture. Note that her measure of labor supply is on the intensive margin, and one way of reconciling these two sets of results is a model in which the extensive margin of women's labor supply in agriculture is more responsive to factors that predate child gender such as the household's landholdings.

²⁶In the sample of all births – where younger parity daughters face differentially higher mortality than boys ([Gupta, 1987](#)) – the pattern of results by age could be affected by the fact that women with higher autonomy are differentially likely to report girls, especially those born long ago.

may agree that children of both genders should complete primary school, but differ in whether girls should go to secondary school. Then, if mothers either do not have full information about these divergent preferences or face constraints in saving up for future spending on a daughter, they would wait to exercise their autonomy until the differential preferences emerge. While this explanation is consistent with lower investment in the education of adolescent girls discussed in section 4.2.3 and shown in table A4, we cannot provide conclusive evidence for this hypothesis. In particular, an alternative explanation for this pattern of age effects is that adolescent daughters free up their mothers to enter the labor force (which subsequently leads to increased autonomy), a hypothesis which we consider in section 5.

Because the survey only sampled women up to age 50, the sample of older children does not capture women who had children relatively late. For instance, the sample of children aged 20 and above loses mothers older than age 30 when the child was born; this represents the 85th percentile of age at birth among surveyed women in India and the 88th percentile in Bangladesh. Inasmuch as women who delay childbirth are somewhat more liberal, the effects of child gender at older ages should be interpreted as the effects on a relatively more conservative sample.

4.2 Other Predictions: Father's Labor Supply, Mother's Consumption, and Contribution to the Household Public Good

4.2.1 Father's Labor Supply

The model predicts that fathers decrease their labor supply with daughters because the mother covers a larger percentage of the child's expenses. While labor supply on the extensive margin is almost universal (99% of married fathers in the Bangladesh data and 97% of married fathers in the Indian data work outside the home), we estimate the effects of daughters on labor supply in India including both the extensive and the intensive margin using the 1993 National Sample Survey (NSS), which contains information on the number of days (between 0 and 7) a respondent worked in the previous week. We use a tobit model to estimate analogous equations to 5 and 6; results are qualitatively similar if we use an ordered logit.

$$DaysWorked_i = \beta_1 ChildrenPresent_i + \beta_2 FemaleChildrenPresent_i + x_i' \delta + \epsilon_i \quad (9)$$

$$DaysWorked_i = \beta_3 OldestChildPresentisFemale_i + x_i' \delta + \epsilon_i \quad (10)$$

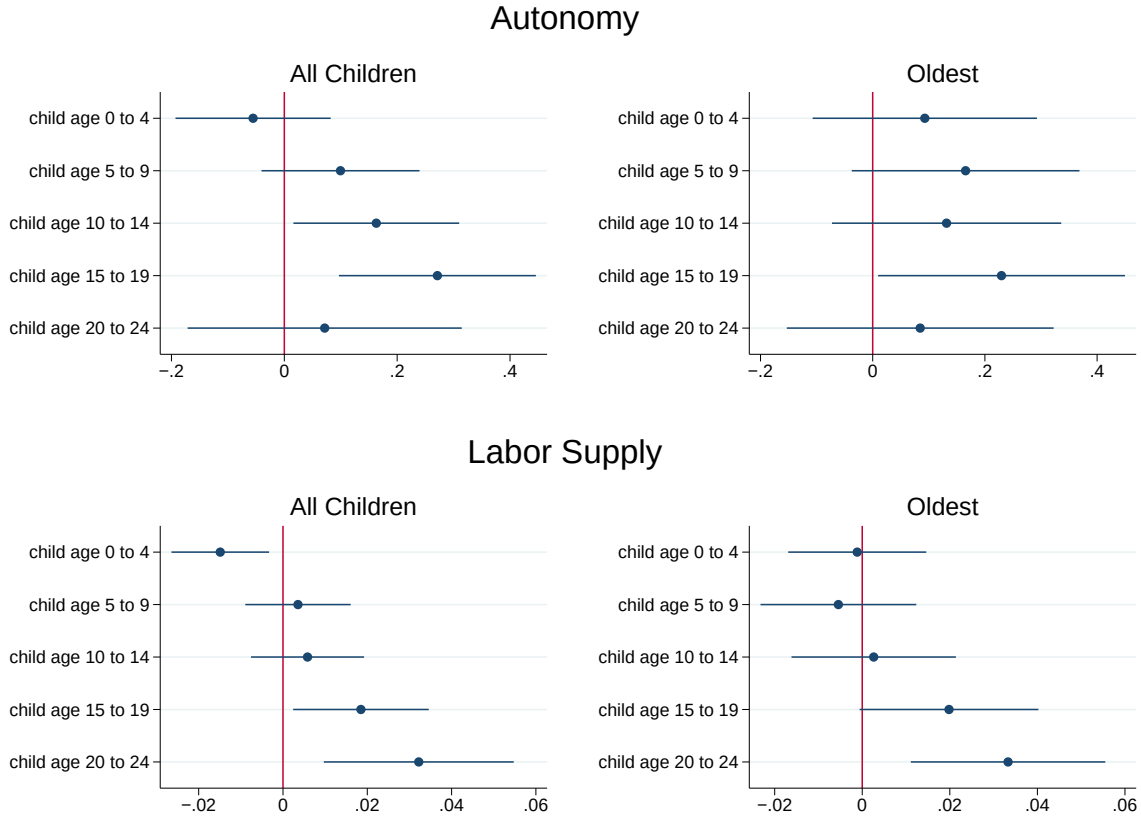


Figure 3: Effects of a daughter on mother's autonomy and labor supply in India, by daughter's age

A caveat to interpreting these results is that rather than using all children ever born to estimate $Children_i$, $Daughters_i$, and $FirstBornDaughter_i$, which we argue in section 3.2 is important to avoid reverse causality concerns, the NSS only records children present in the household. To create a sample of parents and children, we use adults who identify as the head of the household or spouse and consider children listed as children of the head. There are two levels of selection one may be concerned about in this sample: selection into remaining alive and selection into remaining in the household, conditional on being alive.

We try to minimize the latter by considering children who are less likely to be married. However, given that figures 3 and 4 indicate the daughters begin affecting their mothers autonomy and labor supply at approximately age 10, we also would not expect to see a sample of young children generating the results we see. We thus consider maximum age cutoffs ranging from age 10 to 16, the median age at marriage of women interviewed in

		Outcome = Days/week worked															
Panel A: Men																	
daughters	-0.145 [0.1144]	-0.1438 [0.1090]	-0.1598 [0.1025]	-0.1315 [0.0988]	-0.108 [0.0966]	-0.0634 [0.0934]	-0.0971 [0.0906]										
children	0.087 [0.1160]	0.1339 [0.1088]	0.1459 [0.0996]	0.1379 [0.0948]	0.1492 [0.0912]	0.1122 [0.0867]	0.1453* [0.0829]										
first birth female	-0.2648 [0.1729]	-0.2901* [0.1664]	-0.3052* [0.1601]	-0.2751* [0.1561]	-0.2852* [0.1532]	-0.2414 [0.1490]	-0.2658* [0.1453]										
max age oldest child	10	10	11	12	13	14	15	16	16	16	16	16	16	16	16	16	16
Observations	27,134	27,134	29,566	33,817	36,730	39,688	43,552	47,069	47,069	47,069	47,069	47,069	47,069	47,069	47,069	47,069	47,069
Panel B: Women																	
daughters	0.2307 [0.1820]	0.3011* [0.1684]	0.2342 [0.1507]	0.2764* [0.1417]	0.3244** [0.1337]	0.3484*** [0.1259]	0.4059*** [0.1201]										
children	-0.2875* [0.1744]	-0.2868* [0.1597]	-0.2422* [0.1425]	-0.3164** [0.1335]	-0.3095** [0.1253]	-0.3349*** [0.1160]	-0.3550*** [0.1089]										
first birth female	0.3294 [0.2690]	0.3088 [0.2522]	0.2486 [0.2333]	0.4045* [0.2221]	0.4872** [0.2114]	0.6168*** [0.2005]	0.7854*** [0.1922]										
max age oldest child	10	10	11	12	13	14	15	16	16	16	16	16	16	16	16	16	16
Observations	28,141	28,141	30,724	35,245	38,388	41,583	45,770	49,581	49,581	49,581	49,581	49,581	49,581	49,581	49,581	49,581	49,581

Notes: Tobit estimations. Robust standard errors in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Controls not shown include dummies for parent's education level (illiterate/less than primary, literate/less than primary; primary; middle; secondary; undergraduate or graduate; unknown), a dummy for rural location, Hindu religion, state FE's, and dummies for scheduled caste, scheduled tribe, and other backward caste.

Table 3: Effects of Child's Gender on Parents' Labor Supply on the Intensive Margin

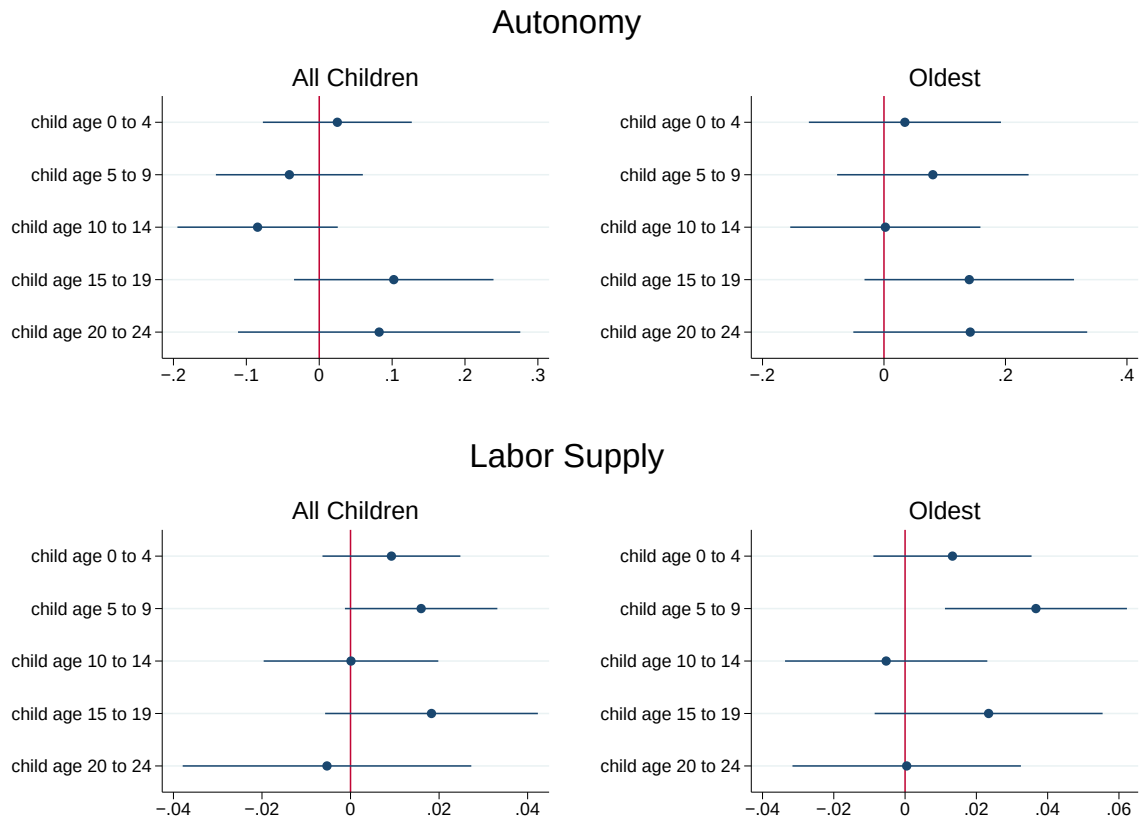


Figure 4: Effects of a daughter on mother's autonomy and labor supply in Bangladesh, by daughter's age

the 1993 National Family Health Survey.

The results, given in table 3, support the model's predictions on both parents' labor supply. Panel A, focusing on men, show that a firstborn daughter decreases a father's days of work by approximately 0.3 days. Panel B shows that the results on women's labor supply on the extensive margin discussed in section 4.1 also hold on the intensive margin. They get particularly strong among mothers of older daughters; having a first-born daughter up to age 16 in the household corresponds to an increase of 0.8 days work per week.

While the mechanism between these results could be a mixture of true causal effects of daughters and selection effects that increase with age (having a 16-year-old daughter still present in the household signifies a greater degree of liberalness than having a ten-year-old daughter still present in the house), note that the most clear selection story suggests that mothers with enough bargaining power to keep her daughter alive and present in

the household may also be more likely to work. By contrast, it is less clear to us which way the selection of having a daughter present in the household would affect men's labor supply. But since we cannot rule out that selection effects are driving the male labor supply results, we view them as suggestive evidence, in accordance with [Rose \(2000\)](#), that support our main model.

4.2.2 Mother's Consumption

The model yields ambiguous predictions on the effect of daughters on their mothers' consumption. On one hand, the mother sacrifices her own consumption to give more to her daughter. On the other hand, once she fights for autonomy (and begins working more) she has greater control over the income she does have.

When interpreting the measures of consumption available in the DHS/NFHS – namely, anemia and BMI – an important consideration is that daughters prompt reoptimization that can affect a mother's health, as reflected in these measures. For instance, daughters tend to prompt higher fertility as their families try again for a son ([Clark 2000](#); [Jensen 2003](#); [Rosenblum 2013](#)). As a results, mothers tend to breastfeed daughters for shorter time than sons in order to be able to get pregnant again more quickly ([Jayachandran and Kuziemko, 2011](#)). The ensuing subsequent pregnancies at close intervals can hurt a mother's health ([Milazzo, 2014](#)). While there is some evidence that breastfeeding can temporarily lead to increased anemia ([Sharman, 2000](#)) – so that mothers of daughters might actually be better nourished in the short run – the ultimate findings of [Milazzo \(2014\)](#) suggest that eventually the effect of subsequent and any other effects of daughters is more harmful and the net effect of daughters on mothers' health is negative. So the effects of daughters on a mother's nutritional status we estimate are a combination of the net effect (illustrated by our model) of the mother increasing her labor supply and seeking more autonomy but needing to divert resources to a girl and any health effects of subsequent fertility that would be reflected in BMI and anemia.

Note, however, that the fact that figures 3 and 4 suggest that autonomy doesn't increase until a daughter is older, while the health effects would operate more when the daughter is younger and her mother is more likely to still be having children. Accordingly, we first provide the overall effects of daughters on their mother's nutrition in table 4, but then we also break them down by age in figure 5. We find, in accordance with the results of [Milazzo \(2014\)](#) in India, that younger daughters tend to hurt their mother's nutritional status in both India and Bangladesh. The effects go away in mothers of older girls, suggesting that the positive effect of increased autonomy on consumption is roughly equal in magnitude to the personal consumptions she sacrifices to increase investment in

Outcome	India		Bangladesh	
	BMI	Anemia	BMI	
daughters	0.0251 [0.0188]	0.0188* [0.0110]	0.0274 [0.0299]	
children	-0.1663*** [0.0174]	0.0084 [0.0094]	-0.2821*** [0.0279]	
first birth female	0.0066 [0.0325]	0.0216 [0.0182]	-0.0088 [0.0510]	
Mean Dep. Var.	21.291	21.291	21.357	21.357
Observations	75,874	75,874	72,423	72,423
R-squared	0.234	0.232	0.163	0.146

Notes: BMI is estimated using OLS and Anemia is estimated with an ordered logit (categories = none, mild, moderate, and severe). Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include mother's education, age fixed effects, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh).

Table 4: Effects of Child's Gender on Mother's Consumption

her daughter, as well as any persistent health effects that began when their daughters were younger.

4.2.3 Contribution to the Household Public Good

The differential preferences of the father for investment in sons predicts that investments will be lower in daughters than in sons. Previous literature has tested for potential differences in parental investments in daughters versus sons. While not every study finds evidence for differential treatment (Duflo, 2005), other studies in South Asia find evidence that parents invest less money (Kingdon 2005; Zimmermann 2012b) and less time (Jayachandran and Kuziemko 2011; Bharadwaj and Lakdawala 2013; Barcellos, Carvalho and Lleras-Muney 2014) in girls than boys, especially after bad economic shocks (Rose, 1999).

Given that most of this literature has focused on young children, we complement this literature by examining educational investments in older children in the India Human Development Survey of 2012, given our finding that differential autonomy appears mostly among older children. As with the tests of labor supply, we consider various age ranges

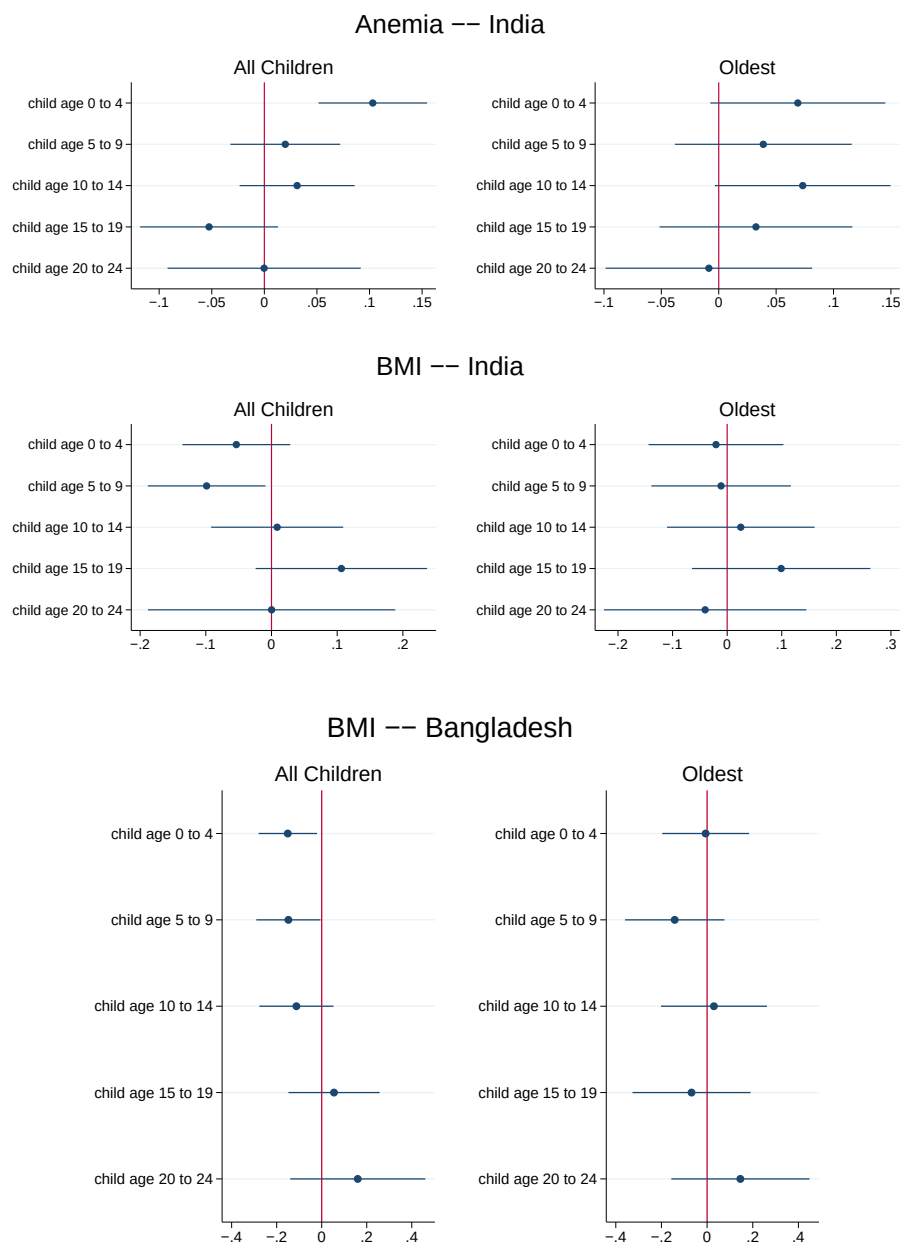


Figure 5: Effects of a daughter on mother's consumption, by daughter's age

of children (which mitigates the concern of endogenous departures from households that we also had when using the NSS survey to examine labor supply). The results, given in [A4](#), show evidence of lower parental education investment in girls. For instance, we find that among firstborn children ages 10 to 12, there is on average 19% less spending on education of daughters than of sons; the difference increases to 32% when we consider children up to age 16.

5 Daughters affect their mothers' labor supply

As we summarized in the extensions to the baseline model mentioned in section [2.1](#) and explained in detail in appendix [B.1](#), previous research has found that daughters have many effects on their families in South Asia that could lead to increased labor supply of mothers.²⁷ In particular, daughters can help with home production, allowing mothers to enter the labor force (as in extension I in appendix [B.1](#)). The need to provide daughters with dowry decreases a household's permanent income, which tends to prompt both parents to increase their labor supply (as in extension II). Relatedly, if the higher future expected transfers from sons are more important to mothers than fathers given the probability they will be widowed ([Lambert and Rossi, 2016](#)), mothers of daughters may work more to compensate for the lack of insurance from sons. Finally, if higher returns to investment in boy's human capital also prompt mothers to invest more time in boys ([Rose, 2000](#)), as in extension IV, mothers may drop out of the labor force to spend more time with their sons, which would imply that daughters are associated with higher maternal labor supply than sons. In any case, working itself could increase a woman's autonomy if it improves her outside option, either through returns to labor market experience or if the act of working itself decreases her psychic cost of working outside the home.

When we incorporate these alternative explanations into our main model in appendix [B.1](#), the prediction that daughters increase their mothers' autonomy persists. In this section, we make a related argument: models based on other effects of daughters – but without allowing the mother to choose to seek greater autonomy after having a daughter – do not naturally explain the autonomy increases we see in section [4](#). That is, while we do not seek to definitively rule out all other ways in which daughters affect their mother's autonomy, we argue that our endogenous autonomy model is likely to be one of the factors.

Some suggestive evidence of the role of these mechanisms comes in table [A5](#), when

²⁷While we focus here on the potential effects of daughters that would directly increase their mothers' labor supply, appendix [C](#) discusses other effects of daughters, namely, on their mother's fertility and subsequent health, or preferences and beliefs.

we break down the effects of a daughter by the sex composition of her siblings. Income effects of daughters of mother's labor supply would likely either be independent of the sex composition of her siblings or increasing in the number of sisters (given decreasing marginal utility of income). Similarly, if the mother has chosen to stay at home with a brother, the effects of a daughter could be less if she has already has brothers. By contrast, if daughters help with housework, the effect of a daughter could be less strong if she has more sisters who can already help out. The results indicate that in India, the effect of a daughter on labor supply is stronger if she has more brothers (either older or younger), consistent with the mechanisms that daughters help with housework. Importantly, however, note that the the lack of heterogeneity in the autonomy effect by sibling composition suggests that labor supply is not the entire channel for the autonomy results, in line with the rest of the result in this section that suggest that even though daughters may bolster their mother's labor supply, this is not the only mechanism behind the labor supply results.

We provide several other tests suggesting that increased labor supply cannot explain the entire autonomy effect. Second, note that increases in labor supply would precede increases in autonomy, while figure 3 suggests that if anything, in India the increases in autonomy come before increases in labor supply. The fact that the strongest labor supply and autonomy effects are seen in mothers of older girls also argues against a tendency for women to drop out of the labor force to spend more time with boys, since the greatest returns to time inputs in children would likely come before they start school. In Bangladesh, shown in figure 4, by contrast, if anything, work may indeed come before labor supply.

To formally test whether the change in autonomy appears before the change in labor supply, we reestimate equation 6 allowing the coefficient on the gender of the firstborn to fully vary based on dummies for the child's age in years. We also run a specification at the child level that examines the effect of the age of all children on the mother's autonomy and labor supply.²⁸ We then estimate the average age driving the autonomy and labor supply effect by estimating the average age of children in the sample, weighted by the estimated treatment effect at each age, in table 5.²⁹ We find that among first children in

²⁸That is, for child i , we estimate

$$\text{Mother's Outcome}_i = \sum_{a=0}^{25} \delta_a 1\{\text{age}_i = a\} + \sum_{a=0}^{25} \beta_a 1\{\text{age}_i = a\} \times \text{daughter}_i + \text{MotherLevelControls}_i + \epsilon_i$$

with standard errors clustered at the mother level. Table A6 provides the results of this regression, as well as the mother-level estimation using only the first-born child.

²⁹That is, we estimate $\text{MeanAge} = \sum_{a=0}^{25} \hat{\beta}_a \times a$ for $\hat{\beta}$'s estimated from both the autonomy and work regressions.

India, the mean age of a daughter driving the autonomy results is 12.1 years, while the mean age of a daughter driving the work results is 14.8 years (P-value of the difference < 0.001). In Bangladesh, the results go in the opposite direction; the mean age of a daughter driving the autonomy results is 14.2 years and the mean age of a daughter driving the labor supply results is 11.8 (P-value of the difference = 0.062). Thus, this test provides some evidence that autonomy comes before labor supply in India, but not Bangladesh.

Third, if increased labor supply is the main channel to explain the increase in female autonomy with a daughter, we might expect the increase to get smaller and insignificant when we focus on women who don't work. Table 6 examines the relationship between daughters and autonomy in this subsample. We emphasize that they should no longer be interpreted as the causal effects of child gender, given that they are estimated on a selective sample that is affected by child gender, given that table 2 demonstrated that daughters increase the probability that their mother works outside the home. To interpret the direction of this selection, recall that our model suggested that the subset of women who don't work (those whose desired hours of work, e^* , are negative) have low baseline autonomy. Given our finding that daughters prompt women to work, women with daughters who do not work presumably have even lower baseline autonomy, so this selection bias would tend to attenuate the relationship between daughters and autonomy in this sample. Nevertheless, table 6 shows a very similar effect (slightly larger in India; slightly smaller in Bangladesh) as in the full sample. As those women do not work outside the home even with a daughter, the increase in their autonomy is unlikely to be driven entirely by a greater labor supply. Instead, consistent with our baseline model, the increased autonomy in women who do not work outside the home, even with a daughter, suggests that at least some of the relationship between daughters and mother's autonomy is driven by a stronger desire to control a fixed amount of income, in light of differential preferences between parents for spending on a daughter.

Fourth, we conduct a similar test that avoids concerns about the endogeneity of the woman's own labor supply by examining heterogeneity based on the average labor supply of other women in a respondent's area, namely, the primary sampling unit of the survey, and her state (India) or region (Bangladesh). The results, depicted in table A7, suggest that if anything, the autonomy results are driven by areas with low female labor force participation. While this fact does not definitively rule out the possibility that labor supply is a channel linking daughters and autonomy – perhaps daughters particularly prompt labor supply in which it is rare for a woman to work – the absence of a differential effect of daughters on autonomy in areas with low female labor force participation argues against this possibility.

Children Included	Firstborn		All	
	Autonomy	Work	Autonomy	Work
Panel A: India				
Average age	12.070 [1.426]	14.576 [1.535]	12.735 [1.446]	15.671 [1.420]
P-value of test: Average age of autonomy effects = average age of work effects				
	0.0115		0.0143	
Panel B: Bangladesh				
Average age	14.120 [1.556]	11.805 [1.444]	14.977 [1.571]	12.291 [1.510]
P-value of test: Average age of autonomy effects = average age of work effects				
	0.0162		0.0134	

*Notes: Average age is the mean age of children in the sample ages 25 and under, weighted by the estimated treatment effect at each age. Standard errors in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The regression that estimates average treatment effect by age includes sampling weights and age of child fixed effects, age of mother fixed effects, mother's education, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh). P-values that estimate the difference based on 100 bootstrap repetitions, where each replicate is stratified at the primary sampling unit level for columns 1-2 and the mother level for columns 3-4. The autonomy index in India is an aggregate of whether the woman has say in large household decisions, purchases for daily life, her own health care, visits to family, and spending her husband's income. Additionally, it includes whether she has access to a bank account and whether she can go alone to the market, the health clinic, and locations out of town. In Bangladesh the autonomy index is an aggregate of whether the woman has say in large household decisions, purchases for daily life, and her own health care. (See section 3.1 for details).*

Table 5: Average Age driving the Autonomy and Labor Supply Effects

Taken together, these tests provide suggestive evidence that tendency for daughters to allow their mothers to work does not explain the entire autonomy results. This is particularly true in India, where the fact that the autonomy results come before the work results provides further evidence of our main model, in which the mother seeks autonomy in order to provide for a daughter. However, we acknowledge that we cannot fully rule out this channel. Even so, the results still highlight that events traditionally viewed as positive for women (here, the birth of a son) may not lead to increases in autonomy as it is commonly conceptualized and measured.

6 Conclusion

While on average mothers in South Asia report a preference for sons over daughters, we show that mothers of daughters have higher participation in household decision-making and freedom of mobility than mothers of sons. These results are consistent with a theoretical model in which mothers have differentially stronger preferences for spending on daughters than do their husbands. This model then predicts that mothers exert greater autonomy, even when doing so is costly for them, and increase their labor supply in order to direct more resources to their daughters. The assumption that seeking autonomy is costly for mothers fits with other papers that show that women's autonomy can provoke violence ([Eswaran and Malhotra 2011](#); [Heath 2014](#); [Anderson and Genicot 2015](#)) or impart psychological costs on the husband ([Bertrand, Kamenica and Pan, 2015](#)). We also show that a possible complementary mechanism to the woman's decision to seek more autonomy is the man's decision to grant more autonomy.

This model suggests new policies that could address son preference. Women could prefer a son even if they get the same fundamental utility from the well-being of a son and a daughter. Thus, policies to address son preference do not necessarily need to address deep-seated social norms biasing women against female offspring. Rather, women need to be compensated for the sacrifice in their own consumption and cost of obtaining autonomy to divert resources to their daughters. And policies that either increase women's baseline autonomy or lower the cost to women of seeking autonomy would also tend to decrease son preference.

Our results also provide new insights into the ways to assess whether women's preferences are taken into account in household decision-making. We argue that low reported levels of autonomy in the way it is typically measured (such as a woman's say in household decision-making) do not necessarily imply that the woman's preferences are ignored, if the household member who is making decisions has similar preferences.

	India		Bangladesh	
	Autonomy index		Autonomy index	
daughters	0.0859**		0.0639***	
	[0.0341]		[0.0235]	
children	-0.036		-0.0822***	
	[0.0314]		[0.0218]	
first birth female		0.1272**		0.1115***
		[0.0585]		[0.0393]
mother's education	0.1412***	0.1405***	0.0279***	0.0324***
	[0.0091]	[0.0090]	[0.0051]	[0.0053]
Mean Dep. Var.	-0.488	-0.488	-0.116	-0.116
Std Dev of Dep. Var.	5.462	5.462	2.528	2.528
Observations	51,687	51,687	20,238	20,238
R-squared	0.195	0.195	0.051	0.046

Notes: Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include a dummy for rural location, age fixed effects, Hindu religion, region FE's (Bangladesh), state FE's (India), survey round (Bangladesh), and dummies for scheduled caste, scheduled tribe, and other backward caste (India). The autonomy index is an aggregate of whether the woman has say in large household decisions, purchases for daily life, her own health care, visits to family, spending her husband's income. Additionally, it includes whether she has access to a bank account and whether she can go alone to the market, the health clinic, and locations out of town. In Bangladesh the autonomy index is an aggregate of whether the woman has say in large household decisions, purchases for daily life, and her own health care. (See section 3.1 for details).

Table 6: Relationship between Child's Gender and Mother's Autonomy among Women who do not Work

Conversely, since fighting for autonomy could have a cost to the woman, high reported values of autonomy may not always imply that the woman's utility is higher. Ultimately, we argue that outcomes such as taking part in household decisions are a means to an end – the woman's preferences are reflected in household decisions – rather than a goal in and of itself.

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APPENDICES FOR ONLINE PUBLICATION

	Autonomy index		1(Work)	
Daughters	0.0669**		0.0053**	
	[0.0264]		[0.0024]	
Daughters X Sex Ratio	0.0011*		0.0001**	
	[0.0006]		[0.0001]	
Children	-0.0919***		-0.0010	
	[0.0263]		[0.0023]	
Children X Sex Ratio	-0.0044***		-0.0001**	
	[0.0005]		[0.0001]	
First Birth Female		0.1157**		0.0085**
		[0.0453]		[0.0042]
First Birth Female X Sex Ratio		0.0004		0.0001
		[0.0010]		[0.0001]
Mother's Education	0.1520***	0.1558***	-0.0124***	-0.0126***
	[0.0076]	[0.0076]	[0.0007]	[0.0007]
Mean Dep. Var.	0.000	0.000	0.351	0.351
Std Dev of Dep. Var.	5.233	5.233	0.477	0.477
Observations	79,827	79,827	79,686	79,686
R-squared	0.200	0.198	0.145	0.145

Notes: Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Sex ratio is number of women for every 1000 men the subtracting the average value across the population (936) so that the main effect of daughters/first birth female gives the effect at the average sex ratio. Data on sex ratio from the 2001 and 2011 censuses (<http://www.census2011.co.in/sexratio.php>). Controls not shown include age fixed effects, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh). The autonomy index is an aggregate of whether the woman has say in large household decisions, purchases for daily life, her own health care, visits to family, and spending her husband's income. Additionally, it includes whether she has access to a bank account and whether she can go alone to the market, the health clinic, and locations out of town. (See section 3.1 for details).

Table A1: Effects of Child Gender on Mother's Autonomy and Labor Supply, Heterogeneity by Sex Ratio of a State

Panel A: India	Variation		Mean Dependent Variable
	All Children	Firstborn	
say in own health care	0.0024 [0.0025]	0.0053 [0.0042]	0.649
say in large household purchases	0.0012 [0.0026]	0.0034 [0.0044]	0.570
say in visits to family	0.0033 [0.0025]	0.0079 [0.0041]	0.649
say in purchases for daily needs	0.0035 [0.0025]	0.0078 [0.0041]	0.634
say in spending husband's earnings	-0.0021 [0.0025]	0.0055 [0.0041]	0.717
can go to market alone	0.0047* [0.0025]	0.0086** [0.0041]	0.576
can go to health facility alone	0.0057 [0.0025]	0.0099** [0.0043]	0.553
can leave village alone	0.0061** [0.0025]	0.0123*** [0.0042]	0.436
has access to a bank account	0.0027 [0.0017]	-0.0048 [0.0030]	0.140
Panel B: Bangladesh			
say in large household purchases	0.0130*** [0.0043]	0.0207*** [0.0071]	0.618
say in visits to family	0.0153*** [0.0041]	0.0149*** [0.0071]	0.608
say in purchases for daily needs	0.0119*** [0.0041]	0.0134*** [0.0067]	0.638

Notes: Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include age fixed effects, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh).

Table A2: Effects of child gender on specific components of the autonomy index

	work for...			work away from		
	self		family	home		
daughters	-0.0007		0.0015	0.0056**		
	[0.0013]		[0.0021]	[0.0023]		
children	0.0000		0.0029	0.0038*		
	[0.0013]		[0.0020]	[0.0023]		
first birth female	-0.0033		0.0041	0.0100**		
	[0.0022]		[0.0036]	[0.0040]		
Mean Dep Var	0.063	0.063	0.205	0.351	0.351	
Observations	79,827	79,827	79,827	79,827	79,827	
R-squared	0.030	0.030	0.135	0.199	0.199	
			0.134		0.199	
			0.086			
	pay scheme...			regularity		
	unpaid		cash	all year		occasionally
daughters	-0.0003		0.0076***	0.0049**	0.0017	
	[0.0017]		[0.0023]	[0.0023]	[0.0021]	
children	0.0016		-0.0048**	-0.0007	0.0019	
	[0.0015]		[0.0021]	[0.0021]	[0.0019]	
first birth female	0.0004		0.0118***	0.0093**	0.0014	
	[0.0027]		[0.0040]	[0.0038]	[0.0035]	
Mean Dep Var	0.104	0.104	0.281	0.257	0.179	0.179
Observations	79,827	79,827	79,827	79,827	79,827	79,827
R-squared	0.100	0.100	0.097	0.079	0.079	0.118

Table A3: Effects of Child's Gender on Mother's Labor Supply, by Sector (India only)

	sector			
	professional /manage/tech	clerical / sales / service	manual	agriculture
daughters	0.0012* [0.0006]	0.0007 [0.0011]	0.0046*** [0.0015]	0.0006 [0.0022]
children	-0.0023*** [0.0005]	-0.0005 [0.0009]	-0.0026** [0.0013]	0.0064*** [0.0022]
first birth female	0.0006 [0.0012]	0.0036* [0.0019]	0.0016 [0.0026]	0.0052 [0.0038]
Mean Dep Var	0.031	0.069	0.084	0.220
Observations	79,827	79,827	79,827	79,827
R-squared	0.091	0.033	0.032	0.262
				0.261

Notes: Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include mother's education, a dummy for rural location, age fixed effects, Hindu religion, state FE's, dummies for scheduled caste, scheduled tribe, and other backward cast.

Effects of Child's Gender on Mother's Labor Supply, by Sector (India only – continued)

		Dependent Variable = Log(1 + Educational Spending)							
Child is Female		-0.3105*** [0.0421]	-0.3233*** [0.0556]	-0.3120*** [0.0452]	-0.3434*** [0.0678]	-0.2745*** [0.0672]	-0.3217*** [0.0844]	-0.1747** [0.0739]	-0.1883* [0.1049]
Oldest Offspring Only?		No	No	No	No	Yes	Yes	Yes	Yes
Ages Included		5-16	10-16	5-12	10-12	5-16	10-16	5-12	10-12
Observations		35,196	22,955	22,059	9,818	11,626	8,048	6,664	3,086
R-squared		0.225	0.171	0.276	0.195	0.232	0.207	0.286	0.246

Notes: Educational Spending is the sum of reported spending on fees (including exam and lab fees); books, uniforms, and other materials; transportation; and private tuition. Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include age fixed effects, religion FE's, and state FE's (India).

Table A4: Effects of Child Gender on Educational Spending in India

Dependent Variable Sample of Children	Auton				Auton			
	index	1(Work)	Auton index	1(Work)	index	1(Work)	Auton index	1(Work)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: India								
Child Female	0.0142 [0.0774]	0.000 [0.0071]	0.0256 [0.0318]	0.0027 [0.0029]	0.0396 [0.0447]	0.0076* [0.0041]	0.1594** [0.0704]	0.0016 [0.0062]
Firstborn Female	0.0845 [0.0624]	0.0120** [0.0058]	0.0976 [0.0676]	0.0166*** [0.0062]				
Child Female X Firstborn Female			-0.0274 [0.0635]	-0.0095* [0.0058]				
Older Siblings					-0.0918*** [0.0272]	-0.0055** [0.0026]		
Child Female X Older Siblings					0.0033 [0.0288]	0.0032 [0.0027]		
Older Sisters					0.0884** [0.0407]	0.0123*** [0.0039]		
Child Female X Older Sisters					0.0007 [0.0409]	-0.0062* [0.0037]		
Younger Siblings							-0.0126 [0.0317]	-0.003 [0.0029]
Child Female X Younger Siblings							-0.0131 [0.0423]	0.0080** [0.0038]
Younger Sisters							0.0385 [0.0458]	0.0080* [0.0042]
Child Female X Younger Sisters							-0.0173 [0.0650]	-0.0104* [0.0059]
Observations	161,533	161,268	161,533	161,268	161,533	161,268	79,331	79,190
R-squared	0.131	0.141	0.131	0.140	0.131	0.141	0.198	0.145

Table A5: Effects of Child Gender on Mother's Autonomy and Labor Supply, Heterogeneity by Sex Composition of Siblings

Dependent Variable Sample of Children	Auton				Auton			
	index	1(Work)	Auton index	1(Work)	index	1(Work)	Auton index	1(Work)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Bangladesh								
Child Female	0.0723 [0.0628]	0.0148 [0.0100]	0.0620** [0.0257]	0.0068 [0.0043]	0.0344 [0.0364]	0.0120** [0.0059]	0.0155 [0.0524]	0.0180** [0.0084]
Firstborn Female			0.1914*** [0.0488]	0.0158** [0.0079]	0.1643*** [0.0545]	0.0209** [0.0087]		
Child Female X Firstborn Female					0.0555 [0.0515]	-0.0103 [0.0085]		
Older Siblings					-0.1331*** [0.0217]	-0.0063* [0.0034]		
Child Female X Older Siblings					0.0004 [0.0231]	-0.0022 [0.0036]		
Older Sisters					0.1478*** [0.0333]	0.0106** [0.0052]		
Child Female X Older Sisters					-0.010 [0.0326]	-0.0009 [0.0052]		
Younger Siblings							-0.1161*** [0.0244]	-0.0162*** [0.0040]
Child Female X Younger Siblings							0.0136 [0.0328]	0.0035 [0.0053]
Younger Sisters							0.0316 [0.0367]	0.0118** [0.0059]
Child Female X Younger Sisters							0.070 [0.0513]	-0.0081 [0.0084]
Observations	48,483	48,477	48,488	48,482	48,488	48,482	24,874	24,871
R-squared	0.026	0.036	0.024	0.036	0.024	0.036	0.051	0.038

Effects of Child Gender on Mother's Autonomy and Labor Supply, Heterogeneity by Sex Composition of Siblings (continued)

Notes: The unit of analysis is the child. Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include mother's education, mother's age fixed effects, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh). The autonomy index is an aggregate of whether the woman has say in large household decisions, purchases for daily life, her own health care, visits to family, and spending her husband's income. Additionally, it includes whether she has access to a bank account and whether she can go alone to the market, the health clinic, and locations out of town. (See section 3.1 for details).

	India				Bangladesh			
	Auton index	1(Work)	Auton index	1(Work)	Auton index	1(Work)	Auton index	1(Work)
Child Female X Child Age = 0	-0.3820*	0.0112	-0.1101	0.0012	0.1338	-0.0302	0.0872	-0.0002
	[0.2214]	[0.0156]	[0.1314]	[0.0103]	[0.1825]	[0.0202]	[0.1088]	[0.0135]
Child Female X Child Age = 1	-0.0347	-0.0126	-0.1677	0.0019	-0.1379	-0.0004	-0.1978*	0.0072
	[0.2278]	[0.0167]	[0.1300]	[0.0110]	[0.1847]	[0.0218]	[0.1071]	[0.0148]
Child Female X Child Age = 2	0.3045	0.0343*	0.2085	-0.0004	0.0329	0.0424	0.1313	0.019
	[0.2296]	[0.0186]	[0.1327]	[0.0112]	[0.1745]	[0.0264]	[0.1073]	[0.0166]
Child Female X Child Age = 3	0.2186	-0.0136	-0.1706	-0.0193*	-0.0034	0.0344	-0.0006	0.0274*
	[0.2359]	[0.0190]	[0.1282]	[0.0112]	[0.1922]	[0.0271]	[0.1029]	[0.0162]
Child Female X Child Age = 4	0.4344*	-0.0273	0.0958	-0.0104	0.1256	0.0141	0.1727*	-0.0034
	[0.2469]	[0.0202]	[0.1293]	[0.0115]	[0.1734]	[0.0269]	[0.1025]	[0.0169]
Child Female X Child Age = 5	0.0477	0.0102	-0.0748	-0.0126	-0.1231	0.0497*	-0.2040*	0.0151
	[0.2293]	[0.0194]	[0.1286]	[0.0113]	[0.1764]	[0.0283]	[0.1052]	[0.0170]
Child Female X Child Age = 6	0.094	0.004	0.124	-0.008	0.008	0.000	0.012	-0.005
	[0.2306]	[0.0187]	[0.1222]	[0.0111]	[0.1901]	[0.0319]	[0.0991]	[0.0170]
Child Female X Child Age = 7	0.268	-0.013	-0.058	-0.015	0.047	0.034	-0.020	0.014
	[0.2340]	[0.0218]	[0.1230]	[0.0112]	[0.1710]	[0.0317]	[0.0952]	[0.0165]
Child Female X Child Age = 8	0.242	-0.020	-0.029	-0.005	0.0481	0.0008	0.135	-0.0068
	[0.2289]	[0.0208]	[0.1199]	[0.0110]	[0.1725]	[0.0286]	[0.0946]	[0.0164]
Child Female X Child Age = 9	0.092	-0.0188	0.0127	0.002	0.4347**	0.1017***	0.1128	0.0344**
	[0.2517]	[0.0215]	[0.1303]	[0.0119]	[0.1866]	[0.0282]	[0.1034]	[0.0172]
Child Female X Child Age = 10	0.4402**	-0.0072	0.2016*	-0.0119	-0.2338	-0.0495	-0.076	-0.0242
	[0.2217]	[0.0214]	[0.1195]	[0.0110]	[0.1733]	[0.0320]	[0.0937]	[0.0166]
Child Female X Child Age = 11	0.1185	0.024	-0.0006	0.0006	-0.0901	0.0315	-0.2000*	0.0138
	[0.2528]	[0.0233]	[0.1293]	[0.0121]	[0.1923]	[0.0319]	[0.1055]	[0.0184]
Child Female X Child Age = 12	-0.0105	-0.0292	-0.0437	-0.0011	0.0791	-0.0115	0.0338	0.0035
	[0.2261]	[0.0204]	[0.1206]	[0.0112]	[0.1746]	[0.0329]	[0.0964]	[0.0173]
Child Female X Child Age = 13	0.1521	0.0184	0.0666	0.0202*	0.2276	-0.0085	-0.0507	0.0108
	[0.2327]	[0.0202]	[0.1237]	[0.0116]	[0.1794]	[0.0325]	[0.1046]	[0.0179]
Child Female X Child Age = 14	-0.0523	0.0102	0.1183	0.0189	0.1088	0.0139	-0.0269	-0.0044
	[0.2440]	[0.0223]	[0.1302]	[0.0122]	[0.1793]	[0.0351]	[0.1031]	[0.0184]
Child Female X Child Age = 15	0.2531	0.0237	0.2194	0.0118	0.1228	0.0234	0.1669	0.0176
	[0.2423]	[0.0225]	[0.1357]	[0.0129]	[0.1882]	[0.0317]	[0.1076]	[0.0179]
Child Female X Child Age = 16	0.0768	0.0035	-0.056	-0.0007	0.4389**	0.0436	0.2219**	0.0337*
	[0.2496]	[0.0234]	[0.1365]	[0.0126]	[0.1808]	[0.0362]	[0.1093]	[0.0188]
Child Female X Child Age = 17	0.5941**	0.0462*	0.4978***	0.0372***	0.0998	0.0364	0.0119	-0.0024
	[0.2549]	[0.0239]	[0.1411]	[0.0133]	[0.2149]	[0.0429]	[0.1148]	[0.0208]
Child Female X Child Age = 18	0.3602	-0.0174	0.2178	0.0065	0.008	0.0517	0.0947	0.0165
	[0.2443]	[0.0224]	[0.1339]	[0.0126]	[0.1842]	[0.0327]	[0.1053]	[0.0173]
Child Female X Child Age = 19	-0.1458	0.0544**	0.0263	0.0288*	0.1026	-0.0443	0.1067	0.0158
	[0.2760]	[0.0261]	[0.1561]	[0.0148]	[0.2184]	[0.0444]	[0.1276]	[0.0217]
Child Female X Child Age = 20	0.1045	0.0417*	0.2580*	0.0294**	0.1032	0.0247	0.3138***	0.0114
	[0.2333]	[0.0226]	[0.1439]	[0.0135]	[0.1924]	[0.0307]	[0.1132]	[0.0182]

Table A6: Effects of Child Gender on Mother's Autonomy and Labor Supply, Heterogeneity by Child's Age

Child Female X Child Age = 21	-0.0616 [0.2920]	0.0451 [0.0275]	0.0161 [0.1728]	0.0370** [0.0159]	-0.2356 [0.2480]	-0.0036 [0.0398]	0.2141 [0.1392]	-0.001 [0.0220]
Child Female X Child Age = 22	0.2936 [0.2652]	-0.02 [0.0246]	0.175 [0.1590]	0.0021 [0.0152]	0.1966 [0.2053]	-0.0394 [0.0334]	0.2456* [0.1295]	-0.0218 [0.0200]
Child Female X Child Age = 23	-0.1462 [0.2893]	0.0146 [0.0270]	0.0165 [0.1840]	0.0141 [0.0172]	0.1139 [0.2242]	-0.0489 [0.0315]	0.2777** [0.1383]	0.0098 [0.0208]
Child Female X Child Age = 24	0.2621 [0.3210]	0.0928*** [0.0290]	0.2153 [0.2013]	0.0326* [0.0188]	0.4556* [0.2451]	0.0619 [0.0447]	0.4070*** [0.1488]	0.0487** [0.0247]
Child Female X Child Age = 25+	-0.1182 [0.1449]	0.0184 [0.0133]	-0.061 [0.1078]	0.0271*** [0.0101]	0.2760*** [0.0867]	0.0291* [0.0171]	0.2145*** [0.0680]	0.0225** [0.0108]
Children Included	First	First	All	All	First	First	All	All
Mean Dep. Var.	-0.008	0.351	-0.004	0.385	0.000	0.191	0.002	0.189
Std Dev of Dep. Var.	0.477	5.233	0.465	5.290	0.393	2.500	0.392	2.496
Observations	79,827	79,686	240,864	240,458	25,028	25,025	73,362	73,353
R-squared	0.199	0.137	0.15	0.131	0.055	0.042	0.033	0.038

Notes: Standard errors in brackets, clustered at the primary sampling unit for columns 1-2 and 5-6 and at the mother level for columns 3-4 and 7-8. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include age of child fixed effects, age of mother fixed effects, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh). The autonomy index in India is an aggregate of whether the woman has say in large household decisions, purchases for daily life, her own health care, visits to family, and spending her husband's income. Additionally, it includes whether she has access to a bank account and whether she can go alone to the market, the health clinic, and locations out of town. In Bangladesh the autonomy index is an aggregate of whether the woman has say in large household decisions, purchases for daily life, and her own health care. (See section 3.1 for details).

	Autonomy index	1(Work)	Autonomy index	1(Work)
Panel A: India				
daughters	0.2584** [0.1069]	0.0108 [0.0097]	0.0851* [0.0502]	0.0146*** [0.0035]
daughters X average FLFP	-0.5197* [0.2688]	-0.0177 [0.0253]	-0.0611 [0.1075]	-0.0282*** [0.0074]
children	0.1087 [0.0922]	-0.0173** [0.0083]	-0.0311 [0.0423]	-0.0322*** [0.0027]
children X average FLFP	-0.3914* [0.2372]	0.0524** [0.0217]	0.0054 [0.0922]	0.0861*** [0.0061]
first birth female	0.4811*** [0.1755]	0.0224 [0.0162]	0.1864** [0.0854]	0.0109* [0.0059]
first birth female X average FLFP	-0.9520** [0.4400]	-0.0372 [0.0415]	-0.1627 [0.1802]	-0.0097 [0.0134]
Geographic unit	state	state	PSU	PSU
Mean Dep. Var.	0.000	0.351	0.000	0.351
Std Dev of Dep. Var.	5.233	0.477	5.233	0.477
Observations	79,686	79,686	79,685	79,685
R-squared	0.195	0.132	0.194	0.257

Table A7: Effects of Child Gender on Mother's Autonomy and Labor Supply, Heterogeneity by Female Labor Force Participation in the Area

	Autonomy index	1(Work)	Autonomy index	1(Work)
Panel B: Bangladesh				
daughters	-0.0316 [0.0738]	0.0278** [0.0108]	0.0950** [0.0378]	0.0178*** [0.0046]
daughters X average FLFP	0.5662 [0.3721]	-0.0778 [0.0561]	-0.0585 [0.1754]	-0.028 [0.0245]
children	0.0249 [0.0546]	-0.0329*** [0.0073]	-0.1035*** [0.0299]	-0.0411*** [0.0036]
children X average FLFP	-0.6845** [0.2673]	0.0501 [0.0400]	-0.0363 [0.1243]	0.0968*** [0.0165]
first birth female		0.3170** [0.1260]	0.1289** [0.0631]	0.0263*** [0.0086]
first birth female X average FLFP		-1.0393* [0.6101]	-0.1337 [0.2660]	-0.0403 [0.0451]
Geographic unit	region	region	PSU	PSU
Mean Dep. Var.	0.000	0.192	0.000	0.192
Std Dev of Dep. Var.	2.500	0.393	2.500	0.393
Observations	25,028	25,025	25,028	25,025
R-squared	0.053	0.102	0.052	0.168

Notes: Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Average FLFP is mean reported labor supply of women in that geographic unit, leaving out the woman responding. Controls not shown include the main effect of the average FLFP in the PSU (columns 5-8), age fixed effects, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh). The autonomy index in India is an aggregate of whether the woman has say in large household decisions, purchases for daily life, her own health care, visits to family, and spending her husband's income. Additionally, it includes whether she has access to a bank account and whether she can go alone to the market, the health clinic, and locations out of town. In Bangladesh the autonomy index is an aggregate of whether the woman has say in large household decisions, purchases for daily life, and her own health care. (See section 3.1 for details).

Effects of Child Gender on Mother's Autonomy and Labor Supply, Heterogeneity by Female Labor Force Participation in the Area (continued)

Panel A: Perceptions of Domestic Violence

	India		Bangladesh	
Outcome:	1(Domestic Violence Ever Acceptable)			
daughters	-0.0044*		-0.0070*	
	[0.0026]		[0.0039]	
children	0.0132***		0.0181***	
	[0.0024]		[0.0033]	
first birth female		-0.0057		-0.0108
		[0.0044]		[0.0067]
Mean Dep. Var.	0.471	0.471	0.320	0.320
Observations	79,827	79,827	25,028	25,028
R-squared	0.107	0.106	0.028	0.026

Panel B: Domestic Violence (India)

	1(Moderate Domestic Violence)		1(Severe Domestic Violence)	
Outcome:				
daughters	-0.0088***		-0.0011	
	[0.0031]		[0.0023]	
children	0.0256***		0.0094***	
	[0.0026]		[0.0021]	
first birth female		-0.0035		0.0016
		[0.0049]		[0.0034]
Mean Dep. Var.	0.319	0.319	0.104	0.104
Observations	59,626	59,626	59,627	59,627
R-squared	0.108	0.104	0.044	0.043

Notes: Standard errors in brackets, clustered at the primary sampling unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sampling weights included. Controls not shown include age fixed effects, dummy for rural location, Hindu religion, region FE's (Bangladesh), state FE's (India), dummies for scheduled caste, scheduled tribe, and other backward caste (India) and survey round (Bangladesh).

Table A8: Effects of Child's Gender on Intimate Partner Violence

A Appendix: Proofs

Proof of Proposition 2.1: The maximization problem can be rewritten as:

$$\max_{e_f, h_f, y_f, a_f} \beta_f \ln(a_f w_f e_f - y_f) + \gamma_f (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_f \ln(1 - e_f - h_f) + \alpha_f \ln(1 - \frac{a_f - \underline{a}}{\underline{a}})$$

$$\max_{e_m, y_m} \beta_m \ln(w_m e_m + (1 - a_f) w_f e_f - y_m) + \gamma_m (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_m \ln(1 - e_m)$$

Take the first-order conditions ($y = y_m + y_f$):

$$[h_f] : \frac{\gamma_f \lambda}{h_f} = \frac{\delta_f}{1 - e_f - h_f} \rightarrow h_f = \frac{\gamma_f \lambda}{\gamma_f \lambda + \delta_f} (1 - e_f)$$

$$[e_f] : \frac{\beta_f a_f w_f}{a_f w_f e_f - y_f} = \frac{\delta_f}{1 - e_f - h_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f}$$

The last equation is obtained by inserting the value of h_f .

$$[e_f, y_f] : \frac{\beta_f a_f w_f}{a_f w_f e_f - y_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f} = \frac{\gamma_f \rho a_f w_f}{y}$$

$$[e_m, y_m] : \frac{\beta_m w_m}{w_m e_m + (1 - a_f) w_f e_f - y_m} = \frac{\delta_m}{1 - e_m} = \frac{\gamma_m \rho w_m}{y}$$

Take the sum of two equations above:

$$\rho(w_f e_f + w_m e_m - y) = (\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m}) y \quad (11)$$

in which

$$e_f = 1 - \frac{\gamma_f \lambda + \delta_f}{\gamma_f \rho a_f w_f} y, \quad e_m = 1 - \frac{\delta_m}{\gamma_m \rho w_m} y$$

Put e_f and e_m into (15):

$$(\rho w_f - \frac{\gamma_f \lambda + \delta_f}{\gamma_f a_f} y + \rho w_m - \frac{\delta_m}{\gamma_m} y - \rho y) = (\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m}) y$$

$$(\frac{\beta_f}{\gamma_f} + \frac{\gamma_f \lambda + \delta_f}{\gamma_f a_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho) y = \rho(w_f + w_m) \quad (12)$$

Lastly, consider a_f ,

$$[a_f] : \frac{\beta_f w_f e_f}{a_f w_f e_f - y_f} = \frac{\alpha_f}{2a - a_f} \rightarrow \frac{\gamma_f \rho w_f e_f}{y} = \frac{\alpha_f}{2a - a_f}$$

Put y and e_f into $[a_f]$,

$$\frac{\gamma_f w_f}{w_f + w_m} \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho \right) = \frac{(\gamma_f \lambda + \delta_f) w_m}{(w_f + w_m) a_f} + \frac{\alpha_f}{2a - a_f} \quad (13)$$

Consider a wider range of a_f , $a_f \in (0, 2a)$, RHS of (13) first decreases and then increases. If there is one interior solution, there must be two solutions of a_f that satisfy (13). The FOC of a_f equals LHS minus RHS of (13), which is first negative, then positive, and then negative. So the utility of the wife must first decreases, then increases and then decreases. The larger solution of a_f is the utility maximizer. We will focus on this solution.

Now a decrease in γ_m increases LHS of (13), and thus increases the solution of a_f . Since

$$\frac{\gamma_f \rho w_f (1 - \frac{\gamma_f \lambda + \delta_f}{\gamma_f \rho a_f w_f} y)}{y} = \frac{\alpha_f}{2a - a_f} \quad (14)$$

as a_f increases, y/a_f decreases, and then e_f increases and h_f decreases. By (12), y/γ_m must increase, so e_m decreases.

By (13), $a_f > \frac{a \sqrt{(\gamma_f \lambda + \delta_f) w_m}}{\sqrt{(\gamma_f \lambda + \delta_f) w_m} + \sqrt{\alpha_f (w_f + w_m)}}$, and a_f increases to $2a$ as γ_m decreases to 0. Thus, when γ_m is below some threshold γ_m^* , a_f is large enough such that equation (14) implies y decreases. Specifically, let $\tilde{a}_f = \frac{a \sqrt{(\gamma_f \lambda + \delta_f) w_m}}{\sqrt{(\gamma_f \lambda + \delta_f) w_m} + \sqrt{\alpha_f (w_f + w_m)}}$ and let $\hat{a}_f$ be the other autonomy that leads a same value to $\frac{\gamma_f \lambda + \delta_f}{\tilde{a}_f} + \frac{\alpha_f}{2a - \tilde{a}_f}$. Clearly, $\hat{a}_f > \tilde{a}_f$. When $a_f^d > \hat{a}_f$, and given $a_f^d > a_f^s > \tilde{a}_f$, we can show that

$$\frac{\gamma_f \lambda + \delta_f}{a_f^d} + \frac{\alpha_f}{2a - a_f^d} > \frac{\gamma_f \lambda + \delta_f}{a_f^s} + \frac{\alpha_f}{2a - a_f^s}$$

Then we can use (13), by setting $a_f = \hat{a}_f$, to compute the threshold γ_m^* . ■

Proof of Proposition 2.2: The maximization problem for a son can be rewritten as:

$$\max_{e_f, h_f} \beta_f \ln(\underline{a} w_f e_f) + \gamma_f (\lambda \ln h_f + \rho \ln y_m) + \delta_f \ln(1 - e_f - h_f)$$

$$\max_{e_m, y_m} \beta_m \ln(w_m e_m + (1 - \underline{a}) w_f e_f - y_m) + \gamma_m (\lambda \ln h_f + \rho \ln y_m) + \delta_m \ln(1 - e_m) - \alpha_m$$

Take the first-order conditions:

$$[h_f] : \frac{\gamma_f \lambda}{h_f} = \frac{\delta_f}{1 - e_f - h_f} \rightarrow h_f = \frac{\gamma_f \lambda}{\gamma_f \lambda + \delta_f} (1 - e_f)$$

$$[e_f] : \frac{\beta_f}{e_f} = \frac{\delta_f}{1 - e_f - h_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f}$$

The last equation is obtained by inserting the value of h_f .

$$e_f = \frac{\beta_f}{\beta_f + \gamma_f \lambda + \delta_f}, h_f = \frac{\gamma_f \lambda}{\beta_f + \gamma_f \lambda + \delta_f}$$

$$[e_m, y_m] : \frac{\beta_m w_m}{w_m e_m + (1 - \underline{a}) w_f e_f - y_m} = \frac{\delta_m}{1 - e_m} = \frac{\gamma_m \rho w_m}{y_m}$$

Thus,

$$e_m = \frac{(\beta_m + \gamma_m \rho) w_m - \delta_m (1 - \underline{a}) w_f e_f}{(\beta_m + \gamma_m \rho + \delta_m) w_m}, y_m = \frac{\gamma_m \rho (w_m + (1 - \underline{a}) w_f e_f)}{\beta_m + \gamma_m \rho + \delta_m}$$

The maximization problem for a daughter can be rewritten as:

$$\max_{e_f, h_f, y_f} \beta_f \ln(\underline{a} w_f e_f - y_f) + \gamma_f (\lambda \ln h_f + \rho \ln y_f) + \delta_f \ln(1 - e_f - h_f)$$

$$\max_{e_m, t_m} \beta_m \ln(w_m e_m + (1 - \underline{a}) w_f e_f - t_m) + \gamma_m (\lambda \ln h_f + \rho \ln(y_f + t_m)) + \delta_m \ln(1 - e_m)$$

Take the first-order conditions ($y = y_f + t_m$):

$$[h_f] : \frac{\gamma_f \lambda}{h_f} = \frac{\delta_f}{1 - e_f - h_f} \rightarrow h_f = \frac{\gamma_f \lambda}{\gamma_f \lambda + \delta_f} (1 - e_f)$$

$$[e_f] : \frac{\beta_f \underline{a} w_f}{\underline{a} w_f e_f - y_f} = \frac{\delta_f}{1 - e_f - h_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f}$$

The last equation is obtained by inserting the value of h_f .

$$[e_f, y_f] : \frac{\beta_f \underline{a} w_f}{\underline{a} w_f e_f - y_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f} = \frac{\gamma_f \rho \underline{a} w_f}{y}$$

$$[e_m, t_m] : \frac{\beta_m w_m}{w_m e_m + (1 - \underline{a}) w_f e_f - t_m} = \frac{\delta_m}{1 - e_m} = \frac{\gamma_m \rho w_m}{y}$$

Take the sum of two equations above:

$$\rho(w_f e_f + w_m e_m - y) = \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m}\right)y \quad (15)$$

in which

$$e_f = 1 - \frac{\gamma_f \lambda + \delta_f}{\gamma_f \rho \underline{a} w_f} y, \quad e_m = 1 - \frac{\delta_m}{\gamma_m \rho w_m} y \quad (16)$$

Put e_f and e_m into (15):

$$\begin{aligned} (\rho w_f - \frac{\gamma_f \lambda + \delta_f}{\gamma_f \underline{a}} y + \rho w_m - \frac{\delta_m}{\gamma_m} y - \rho y) &= \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m}\right)y \\ \left(\frac{\beta_f}{\gamma_f} + \frac{\gamma_f \lambda + \delta_f}{\gamma_f \underline{a}} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho\right)y &= \rho(w_f + w_m) \end{aligned} \quad (17)$$

It is easier to start from the result with a daughter, and consider when γ_m increases. By (17), y increases and y/γ_m decreases. Then by (16), e_f decreases and e_m increases. This change continues until $y_f = 0$. When it transits to the case with a son, in which only the husband takes care of the household public good, the spouses' problem are completely separate. The wife behavior as if y_f , and thus e_f decreases and h_f increases. She gets less autonomy because it changes from $t_m > 0$ to $t_m = 0$. As r_m further increases, the husband works more and spend more on the child.

In other words, when it is a girl (compared to a son), for the wife, e_f increases, h_f decreases and a_f increases. In addition, e_m decreases and y decreases. ■

Proof of Proposition 2.3: The model with unearned income is discussed in appendix B.3, in particular, (28):

$$\frac{\gamma_f(w_f + R_f)}{w_f + w_m + R_f + R_m} \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho\right) = \frac{(\gamma_f \lambda + \delta_f)(w_m + R_m)}{(w_f + w_m + R_f + R_m)a_f} + \frac{\alpha_f}{2\underline{a} - a_f} \quad (18)$$

As $\underline{a}$ increases, the RHS decreases. Since we focus on the larger solution of a_f , the RHS increases in a_f , and thus an increase in $\underline{a}$ increases the solution of a_f . By (29), y/a_f increases, so e_f increases. ■

Proof of Proposition 2.4: Take the first-order conditions ($y = y_m + y_f$):

$$[h_f] : \quad \frac{\gamma_f \lambda}{h_f} = \frac{\delta_f}{1 - h_f}$$

$$[a_f, y_f] : \frac{\beta_f R_f}{a_f R_f - y_f} = \frac{\gamma_f \rho R_f}{y} = \frac{\alpha_f}{2\underline{a} - a_f}$$

$$[e_m, y_m] : \frac{\beta_m w_m}{w_m e_m + (1 - a_f) R_f - y_m} = \frac{\delta_m}{1 - e_m} = \frac{\gamma_m \rho w_m}{y}$$

Notice that h_f is independent of the rest of the outcomes. So it remains the same in both models.

Sum $[a_f, y_f]$ and $[e_m, y_m]$,

$$\rho(R_f + w_m e_m - y) = \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m}\right)y$$

in which $e_m = 1 - \frac{\delta_m}{\gamma_m \rho w_m} y$. So

$$\left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho\right)y = \rho(w_m + R_f)$$

In both models, as γ_m decreases, y decreases and y/γ_m increases, so e_m decreases and a_f increases. ■

B Appendix: Robustness of the Model

B.1 More Alternative Models

Extension I: daughters help with housework.

One possible difference between daughters and sons is that daughters could help their mothers with some housework. In particular, suppose they contribute H^d hours to produce the household public good. Alternatively, this parameter can reflect daughters' ability to take care of themselves, thereby requiring less time from their mother.

The wife's optimization problem can be summarized as follows ($H^s = 0$ if it is a son),

$$\max_{x_f, h_f, e_f, a_f} \beta_f \ln x_f + \gamma_f (\lambda \ln(h_f + H^{s,d}) + \rho \ln(y_f + y_m)) + \delta_f \ln(1 - e_f - h_f) + \alpha_f \ln\left(1 - \frac{a_f - \underline{a}}{\underline{a}}\right)$$

$$\text{s.t. } e_f, h_f \in [0, 1], a_f \in [\underline{a}, 2\underline{a}], x_f \geq 0, p_f x_f \leq a_f (w_f e_f)$$

Proposition B.1. *Suppose the Nash equilibrium is interior. If daughter helps with housework, i.e. the total time on housework is $h_f + H^d$ with a daughter, then when the child is a daughter (relative to son):*

- The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).
- The husband works less outside (e_m).
- The monetary investment to the public good increases ($y_f + y_m$).

Proof of Proposition B.1: Rewrite the first-order conditions:

$$[h_f] : \frac{\gamma_f \lambda}{h_f + H} = \frac{\delta_f}{1 - e_f - h_f} \rightarrow h_f + H = \frac{\gamma_f \lambda}{\gamma_f \lambda + \delta_f} (1 - e_f + H)$$

$$[e_f] : \frac{\beta_f a_f w_f}{a_f w_f e_f - y_f} = \frac{\delta_f}{1 - e_f - h_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f + H}$$

$$[e_f, y_f] : \frac{\beta_f a_f w_f}{a_f w_f e_f - y_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f + H} = \frac{\gamma_f \rho a_f w_f}{y}$$

$$e_f = 1 + H - \frac{\gamma_f \lambda + \delta_f}{\gamma_f \rho a_f w_f} y$$

Then

$$\left(\frac{\beta_f}{\gamma_f} + \frac{\gamma_f \lambda + \delta_f}{\gamma_f a_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho \right) y = \rho (w_f (1 + H) + w_m) \quad (19)$$

$$\frac{\gamma_f w_f (1 + H)}{w_f (1 + H) + w_m} \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho \right) = \frac{(\gamma_f \lambda + \delta_f) w_m}{(w_f (1 + H) + w_m) a_f} + \frac{\alpha_f}{2a - a_f} \quad (20)$$

With a positive H , LHS of (20) increases and RHS decreases, so a_f has to increase. Note that since RHS decreases and a_f is in the right half of the U shape without H , a_f is also in the right half of the U shape with H . By (19), as a_f and H increase, y must increase. Then e_m decreases. By (15), as y increases and e_m decreases, e_f must increase and then h_f decreases. ■

Since the daughter frees the wife from housework, the wife can work more outside the home and less at home. As she earns more, it is beneficial to get a higher autonomy. With the wife working more and contributing more to the household public good, the husband can work less and enjoy more leisure.

Extension II: daughters cost more – dowry

The second difference between sons and daughters is the need to provide daughters with dowry, which decreases a household's permanent income and could prompt both parents to work more (Deolalikar and Rose, 1998). In a static model, we assume that the household needs to save an exogenous amount D^d for the daughter's dowry. So after paying

the dowry, the total contribution to the household public good is $y_f + y_m - D^d$, while we set $D^s = 0$. Because of the aforementioned reinforcement between female labor supply and her autonomy, a small dowry could in theory push the wife to work more and earn more than covering the dowry, such that the total investment to the child could increase. Since we do not think this is empirically relevant, we introduce a threshold D^* and focus on when $D^d > D^*$ to rule this out.

Proposition B.2. *Suppose the Nash equilibrium is interior. If daughter requires dowry, i.e. the monetary investment in a daughter is $y_m + y_f - D^d$, then when the child is a daughter (relative to son):*

- The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).
- The husband works more outside (e_m), if $D^d \geq D^*$.
- The monetary investment to the public good decreases ($y_f + y_m - D^d$), if $D^d \geq D^*$.

Proof of Proposition B.2: Rewrite the first-order conditions ($y' = y_f + y_m - D$):

$$[e_f, y_f] : \frac{\beta_f a_f w_f}{a_f w_f e_f - y_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f} = \frac{\gamma_f \rho a_f w_f}{y'}$$

$$[e_m, y_m] : \frac{\beta_m w_m}{w_m e_m + (1 - a_f) w_f e_f - y_m} = \frac{\delta_m}{1 - e_m} = \frac{\gamma_m \rho w_m}{y'}$$

$$(w_f e_f + w_m e_m - y' - D) = \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} \right) y'$$

$$\left(\frac{\beta_f}{\gamma_f} + \frac{\gamma_f \lambda + \delta_f}{\gamma_f a_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho \right) y' = \rho (w_f + w_m - D) \quad (21)$$

$$\frac{\gamma_f w_f}{w_f + w_m - D} \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho \right) = \frac{(\gamma_f \lambda + \delta_f)(w_m - D)}{(w_f + w_m - D) a_f} + \frac{\alpha_f}{2a - a_f} \quad (22)$$

$$\frac{\gamma_f \rho w_f \left(1 - \frac{\gamma_f \lambda + \delta_f}{\gamma_f \rho a_f w_f} y' \right)}{y'} = \frac{\alpha_f}{2a - a_f} \quad (23)$$

With a positive D , LHS of (22) increases and RHS decreases, so a_f has to increase. Note that since RHS decreases and a_f is in the right half of the U shape without D , a_f is also in the right half of the U shape with D . By (23), y' / a_f decreases, and then e_f increases and h_f decreases. Similarly, by (22), $a_f > \frac{a \sqrt{(\gamma_f \lambda + \delta_f)(w_m - D)}}{\sqrt{(\gamma_f \lambda + \delta_f)(w_m - D)} + \sqrt{\alpha_f (w_f + w_m - D)}}$, and a_f increases to $2a$ as D increases. Thus, when D is above some threshold D^* , a_f is large enough such that equation (23) implies y' decreases and then e_m increases. ■

Since the daughter brings a negative income shock, both spouses have to work more to pay for the dowry. The wife fights for a higher autonomy because her increased labor supply raises the marginal gains to autonomy, and because the lump-sum dowry taken out of the contribution to the household public good raises the marginal returns to investment in the daughter (since the production of child quality is concave).

Extension III: daughters lower the cost of autonomy.

The husband's preference of a son over a daughter may have additional effects on his behavior, beyond the effects on his labor supply and contribution to the household public good in the baseline model. For instance, he could pay more attention to supervising the wife when he has a son, in order to make sure the investments in the child more closely align with his preferences. One easy way to incorporate this story is to assume that the cost of autonomy to the wife is lower with a daughter than a son, $\alpha_f^s > \alpha_f^d$.³⁰

Proposition B.3. *Suppose the Nash equilibrium is interior. If the cost of autonomy is smaller with a daughter than a son ($\alpha_f^s > \alpha_f^d$), then when the child is a daughter (relative to son):*

- *The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).*
- *The husband works less outside (e_m).*
- *The monetary investment to the public good increases ($y_f + y_m$).*

Proof of Proposition B.3: A decrease in α_f decreases RHS of (13). Since we focus on the larger solution of a_f , the RHS increases in a_f , and thus a decrease in α_f increases the solution of a_f . By (14), as a_f increases, y/a_f decreases, so e_f increases and h_f decreases. By (13) and (14),

$$\gamma_f \rho w_f = \left(\frac{\gamma_f w_f}{w_f + w_m} \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho \right) + \frac{(\gamma_f \lambda + \delta_f) w_f}{(w_f + w_m) a_f} \right) y$$

As a_f increases, y must increase, and then e_m decreases. ■

The lower cost of autonomy with a daughter prompts the wife to seek a higher autonomy, which makes working outside more profitable, and as a result she works more outside and less at home. As the wife brings more money back from working, the husband gets to work less. But the household's total income still increases, as does the monetary

³⁰The husband's response could also be modeled in a principle-agent framework, as in [Eswaran and Malhotra \(2011\)](#). The result would be very similar; a husband would devote less costly monitoring effort to his wife after a daughter than a son, allowing the wife to gain more autonomy.

investment to the public good. So allowing the husband to respond to a son by behaving in a way that makes it more costly for the wife to seek autonomy reinforces our main prediction that wives seek more autonomy after daughters.

Extension IV: daughters give lower return to investment.

It is possible that the income effect of daughters is of a different form than the lump-sum dowry cost in extension II. Specifically, consider the possibility that there are lower returns to parents to investment in a daughter compared to a son (Rosenzweig and Schultz 1982; Rose 2000), either because the labor market returns to girls' human capital are lower or parents value returns less (even if they are equal) if they anticipate lower transfers from daughters after marriage. Specifically, we assume the return to financial contribution ($\rho^{s,d}$) and/or to time contribution ($\lambda^{s,d}$) depends on the child's gender, and investments in daughters give a lower return: $\rho^s > \rho^d$ and/or $\lambda^s > \lambda^d$.

$$z = f(h_f, y_f + y_m) = (h_f)^{\lambda^{s,d}} (y_f + y_m)^{\rho^{s,d}}$$

Proposition B.4. *Suppose the Nash equilibrium is interior. If the return to monetary investment decreases when it is a girl ($\rho^s > \rho^d$),*

- *The wife works less outside (e_f), more at home (h_f), and gets a lower autonomy (a_f).*
- *The monetary investment to the public good decreases ($y_f + y_m$).*

Proof of Proposition B.4: A decrease in ρ decreases LHS of (13), and thus decreases the solution of a_f . By (14), as a_f decreases, $y/(\rho a_f)$ increases, and then e_f decreases and h_f increases. By (12), a_f and ρ decreases, so y decreases. ■

As the return to monetary investment is lower with a daughter, the wife works less outside and more at home. She needs a lower autonomy because she doesn't need to spend much on the household public good and she works less. Thus, the total financial contribution into the public good goes down. The effect on the husband's labor supply is ambiguous because the drop of the wife's labor supply may be too much, directly due to the lower return to investment and indirectly due to the drop of her autonomy. While the husband also wants to work less because of the lower return to investment, he may need to work more if the drop from the wife's side is too much.

Proposition B.5. *Suppose the Nash equilibrium is interior. If the return to time investment in the public good decreases when it is a girl ($\lambda^s > \lambda^d$),*

- *The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).*

- The husband works less outside (e_m).
- The monetary investment to the public good increases ($y_f + y_m$).

Proof of Proposition B.5: A decrease in λ decreases RHS of (13). Since we focus on the larger solution of a_f , the RHS increases in a_f , and thus a decrease in λ increases the solution of a_f . By (13) and (14),

$$\gamma_f \rho w_f = \left(\frac{\gamma_f w_f}{w_f + w_m} \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho \right) + \frac{(\gamma_f \lambda + \delta_f) w_f}{(w_f + w_m) a_f} \right) y$$

As λ decreases and a_f increases, y must increase. By $[a_f]$, e_f increases, by $[h_f]$, h_f decreases, and by $[e_m, y_m]$, e_m decreases. ■

As the return to the time investment is lower with a daughter, the wife works less at home and more outside. She fights for a higher autonomy because she works more. Since the wife earns more, she contributes more to the household public good, which allows the husband to contribute less and thus works less. But the decrease in the husband's contribution will not completely offset the increase from the wife's contribution, so the monetary investment to the household public good increases.

B.2 Autonomy as control of time

Alternatively, we can think of autonomy as the wife's control of her time. In many traditional and conservative societies, the husband may dislike his wife working outside (Basu, 2006). In a simple model, we assume the husband controls h_f , the time the wife has to spend on the household public good (i.e. taking care of the child). And the wife allocates her free time $1 - h_f$ to working outside and leisure. Her autonomy is her free time $a_f = 1 - h_f$.

In the husband's utility,

$$\max_{x_m, h_f, e_m, y_m} \beta_m \ln x_m + \gamma_m^{s,d} (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_m \ln(1 - e_m) + \alpha_m \ln(1 - h_f)$$

$$\text{s.t. } e_m, h_f \in [0, 1], x_m, y_m > 0, p_m x_m \leq w_m e_m - y_m$$

where $\alpha_m \ln(1 - h_f)$ is the cost for the husband to supervise the wife.

In the wife's utility,

$$\max_{x_f, e_f, y_f} \beta_f \ln x_f + \gamma_f (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_f \ln(1 - e_f - h_f)$$

$$s.t. e_f \in [0, 1 - h_f], x_f, y_f > 0, p_f x_f \leq w_f e_f - y_f$$

Proposition B.6. Suppose the autonomy is control of time and Nash equilibrium is interior. If the husband's preference for the public good decreases when it is a girl ($\gamma_m^s > \gamma_m^d$),

- The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).
- The husband works less outside (e_m).
- The monetary investment to the household public good may increase or decrease ($y_f + y_m$).

Proof of Proposition B.6: Take first-order conditions:

$$[h_f] : \frac{\lambda \gamma_m}{h_f} - \frac{\alpha_m}{1 - h_f} = 0 \rightarrow h_f = \frac{\lambda \gamma_m}{\lambda \gamma_m + \alpha_m}$$

So as γ_m decreases when it is a girl, h_f decreases and $a_f = 1 - h_f$ increases.

$$[e_f, y_f] : \frac{\beta_f w_f}{w_f e_f - y_f} = \frac{\delta_f}{1 - e_f - h_f} = \frac{\gamma_f \rho w_f}{y}$$

$$[e_m, y_m] : \frac{\beta_m w_m}{w_m e_m - y_m} = \frac{\delta_m}{1 - e_m} = \frac{\gamma_m \rho w_m}{y}$$

So we have,

$$\rho(w_f e_f + w_m e_m - y) = \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m}\right)y \quad (24)$$

$$e_f = 1 - h_f - \frac{\delta_f}{\gamma_f \rho w_f} y, \quad e_m = 1 - \frac{\delta_m}{\gamma_m \rho w_m} y \quad (25)$$

Putting e_f and e_m into the equation (24),

$$\rho(w_f(1 - h_f) + w_m) = \left(\frac{\beta_f + \delta_f}{\gamma_f} + \frac{\beta_m + \delta_m}{\gamma_m} + \rho\right)y$$

As h_f decreases, y/γ_m must increase, so e_m decreases. Lastly, if y decreases, by the expression of e_f in (25), it increases; otherwise if y increases, by equation (24), e_f must increase. So regardless, e_f increases. ■

As the husband cares less about a girl, he demands less control of the wife's time, so she works less at home, works more outside and gets a higher autonomy. With both the less care of the household public good and the wife's higher earning, the husband works less. The monetary contribution to the household public good is ambiguous, since on one

hand the husband contributes less because he cares less, while on the other hand, the wife contributes more because she earns more.

B.3 Unearned Income

Let R_f be the wife's unearned income and R_m be the husband's unearned income.

The wife's optimization problem can be summarized as follows,

$$\begin{aligned} \max_{x_f, h_f, y_f, e_f, a_f} \quad & \beta_f \ln x_f + \gamma_f (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_f \ln(1 - e_f - h_f) + \alpha_f \ln\left(1 - \frac{a_f - \underline{a}}{\underline{a}}\right) \\ \text{s.t.} \quad & e_f, h_f \in [0, 1], a_f \in [\underline{a}, 2\underline{a}] \quad x_f, y_f \geq 0, \quad p_f x_f + y_f \leq a_f(w_f e_f + R_f) \end{aligned}$$

The husband's optimization problem is similar,

$$\begin{aligned} \max_{x_m, y_m, e_m} \quad & \beta_m \ln x_m + \gamma_m^{s,d} (\lambda \ln h_f + \rho \ln(y_f + y_m)) + \delta_m \ln(1 - e_m) \\ \text{s.t.} \quad & e_m \in [0, 1], \quad x_m, y_m \geq 0, \quad p_m x_m + y_m \leq w_m e_m + R_m + (1 - a_f)(w_f e_f + R_f) \end{aligned}$$

Proposition B.7. *The predictions of Proposition 2.1 remains the same except with a possibly different threshold γ_m^{**} : suppose the Nash equilibrium is interior and the husband's preference for the public good decreases when it is a girl ($\gamma_m^s > \gamma_m^d$),*

- *The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).*
- *The husband works less outside (e_m).*
- *The monetary investment to the household public good decreases ($y_f + y_m$), if $\gamma_m^d \leq \gamma_m^{**}$.*

Proof of Proposition B.7: The proof is similar to the one without unearned income. Take the first-order conditions ($y = y_m + y_f$):

$$\begin{aligned} [h_f] : \quad & \frac{\gamma_f \lambda}{h_f} = \frac{\delta_f}{1 - e_f - h_f} \rightarrow h_f = \frac{\gamma_f \lambda}{\gamma_f \lambda + \delta_f} (1 - e_f) \\ [e_f] : \quad & \frac{\beta_f a_f w_f}{a_f(w_f e_f + R_f) - y_f} = \frac{\delta_f}{1 - e_f - h_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f} \end{aligned}$$

The last equation is obtained by inserting the value of h_f .

$$[e_f, y_f] : \quad \frac{\beta_f a_f w_f}{a_f(w_f e_f + R_f) - y_f} = \frac{\gamma_f \lambda + \delta_f}{1 - e_f} = \frac{\gamma_f \rho a_f w_f}{y}$$

$$[e_m, y_m] : \frac{\beta_m w_m}{w_m e_m + R_m + (1 - a_f)(w_f e_f + R_f) - y_m} = \frac{\delta_m}{1 - e_m} = \frac{\gamma_m \rho w_m}{y}$$

Take the sum of two equations above:

$$\rho(w_f e_f + R_f + w_m e_m + R_m - y) = \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m}\right)y \quad (26)$$

in which

$$e_f = 1 - \frac{\gamma_f \lambda + \delta_f}{\gamma_f \rho a_f w_f} y, \quad e_m = 1 - \frac{\delta_m}{\gamma_m \rho w_m} y$$

Put e_f and e_m into (26):

$$\begin{aligned} (\rho w_f + \rho R_f - \frac{\gamma_f \lambda + \delta_f}{\gamma_f a_f} y + \rho w_m + \rho R_m - \frac{\delta_m}{\gamma_m} y - \rho y) &= \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m}\right)y \\ \left(\frac{\beta_f}{\gamma_f} + \frac{\gamma_f \lambda + \delta_f}{\gamma_f a_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho\right)y &= \rho(w_f + w_m + R_f + R_m) \end{aligned} \quad (27)$$

Lastly, consider a_f ,

$$[a_f] : \frac{\beta_f(w_f e_f + R_f)}{a_f(w_f e_f + R_f) - y_f} = \frac{\alpha_f}{1 - a_f} \rightarrow \frac{\gamma_f \rho(w_f e_f + R_f)}{y} = \frac{\alpha_f}{2\underline{a} - a_f}$$

Put y and e_f into $[a_f]$,

$$\frac{\gamma_f(w_f + R_f)}{w_f + w_m + R_f + R_m} \left(\frac{\beta_f}{\gamma_f} + \frac{\beta_m}{\gamma_m} + \frac{\delta_m}{\gamma_m} + \rho\right) = \frac{(\gamma_f \lambda + \delta_f)(w_m + R_m)}{(w_f + w_m + R_f + R_m)a_f} + \frac{\alpha_f}{2\underline{a} - a_f} \quad (28)$$

As discussed in the proof of Proposition 2.1, the larger solution of a_f is the utility maximizer and we will focus on this solution.

Now a decrease in γ_m increases LHS of (28), and thus increases the solution of a_f . Since

$$\frac{\gamma_f \rho w_f \left(1 - \frac{\gamma_f \lambda + \delta_f}{\gamma_f \rho a_f w_f} y\right) + \gamma_f \rho R_f}{y} = \frac{\alpha_f}{2\underline{a} - a_f} \quad (29)$$

as a_f increases, y/a_f decreases, and then e_f increases and h_f decreases. By (27), y/γ_m must increase, so e_m decreases. The threshold γ_m^{**} can be found using exactly the same method, such that as $\gamma_m^d \leq \gamma_m^{**}$, y decreases. ■

B.4 Corner solutions

Since the vast majority of men in South Asia work outside the home – 97% of married men with children in our data work in India, and 98% of in Bangladesh – we maintain the assumption that the husband’s wage is high enough that he always works outside the home in equilibrium, i.e. $e_m > 0$.³¹

We focus on the corner solution in which only one spouse is the financial contributor of the child’s expense. This is a generalization of models in which the wife contributes time and the husband makes a financial contribution (Anderson and Eswaran 2009). In our model this outcome will result when the husband has a high preference for the child, likely a son. However, when the child is a daughter and the husband cares very little about her well-being, the husband has little incentive to spend money on her and so the wife also takes care of her financial expenses, e.g. her education. Qualitative evidence from Bangladesh suggests that many women do tend to invest money (not just time) resources on their children, and their daughters in particular (Kabeer 1997; Sultan Ahmed and Bould 2004).

Lemma B.8. *There exist $\bar{\gamma}_m$ and $\underline{\gamma}_m$, such that*

- *If the husband cares sufficiently high for the child’s well-being ($\gamma_m \geq \bar{\gamma}_m$), there is an equilibrium in which he pays for the child’s entire expense $y_m > 0$ and $y_f = 0$.*
- *If the husband cares sufficiently little for the child’s well-being ($\gamma_m \leq \underline{\gamma}_m$), there is an equilibrium in which the wife pays for the child’s entire expense $y_f > 0$ and $y_m = 0$.*

Proof of Lemma B.8: When it is a son, the wife has no incentive to invest in the child only when the marginal utility is higher to invest to her own consumption (the equilibrium calculation is in the proof of Proposition B.9):

$$\frac{\beta_f}{x_f^s} \geq \frac{\gamma_f \rho}{y_m^s} \rightarrow \frac{\gamma_m^s \rho}{\beta_m + \gamma_m^s \rho + \delta_m} \geq \frac{x_f^s \gamma_f \rho}{\beta_f (w_m + (1 - a_f^s)(w_f e_f^s))}$$

So there exists an upper bound $\bar{\gamma}_m$, such that when $\gamma_m^s \geq \bar{\gamma}_m$, the husband pays high enough on the son and the wife doesn’t find it optimal to spend on the son.

³¹A corner solution ($e_m = 0$) is not impossible because the husband gets some unearned income from the wife, but happens only if men’s salary is relatively low compared to women’s salary. While our data does not have earnings, in India women who report earnings earned on average 57 % as much as men (2004 census) and in Bangladesh women who reported earnings earned on average 69 % as men (2005 Housing Income and Expenditure Survey), so this case does not appear to be empirically relevant.

When it is a daughter, the husband have no incentives to invest in the child only when the marginal utility is higher to invest in his own consumption:

$$\frac{\beta_m}{x_m^d} \geq \frac{\gamma_m^d \rho}{y_f^d} \rightarrow \gamma_m^d \leq \frac{\beta_m y_f^d}{\rho x_m^d}$$

So there exists a lower bound $\underline{\gamma}_m$, such that when $\gamma_m^d \leq \underline{\gamma}_m$, the husband cares so little about the child that he doesn't want to contribute when the wife invest into the child. ■

We remark that $\bar{\gamma}_m$ and $\underline{\gamma}_m$ depends on all the parameters, including γ_f , and we omit this dependence to simplify the notation. We denote the equilibria above as single-financial-contributor equilibria, with the first type as men-contributing equilibria and the second one as women-contributing equilibria.

To study the effect of the child's gender, we consider a husband with high preference for a son ($\gamma_m^s \geq \bar{\gamma}_m$) and low preference for a daughter ($\gamma_m^d \leq \underline{\gamma}_m$). In other words, the change is from a men-contributing equilibrium for a son to a women-contributing equilibrium for a daughter. While it is not entirely impossible to have multiple equilibria, especially when $\underline{\gamma}_m > \bar{\gamma}_m$, we further restrict our attention to the case with unique equilibrium, such that $\gamma_m^s \geq \max(\bar{\gamma}_m, \underline{\gamma}_m)$ and $\gamma_m^d \leq \min(\bar{\gamma}_m, \underline{\gamma}_m)$.

Proposition B.9. *Suppose $\gamma_m^s \geq \max(\bar{\gamma}_m, \underline{\gamma}_m)$ and $\gamma_m^d \leq \min(\bar{\gamma}_m, \underline{\gamma}_m)$. When the child is a daughter (compared to a son):*

- *The wife works more outside the home (e_f), less at home (h_f), gets a higher autonomy (a_f), but a lower overall utility (u_f).*
- *The husband works less (e_m) and consumes more (x_m) as long as the loss of control of his wife's income is not too high.*

Proof of Proposition B.9: When it is a son, the husband pays the child's entire expense. Then in equilibrium,

$$\frac{\beta_f}{a_f} = \frac{\alpha_f}{2\underline{a} - a_f} \rightarrow a_f^s = \frac{2\underline{a}\beta_f}{\alpha_f + \beta_f}$$

$$\frac{\beta_f w_f}{w_f e_f} = \frac{\delta_f}{1 - e_f - h_f}, \quad \frac{\gamma_f \lambda}{h_f} = \frac{\delta_f}{1 - e_f - h_f}$$

So,

$$e_f^s = \frac{\beta_f}{\beta_f + \gamma_f \lambda + \delta_f}, \quad h_f^s = \frac{\gamma_f \lambda}{\beta_f + \gamma_f \lambda + \delta_f}$$

The husband:

$$y_m^s = \frac{\gamma_m^s \rho}{\beta_m + \gamma_m^s \rho} (w_m e_m + (1 - a_f)(w_f e_f))$$

$$e_m^s = \frac{(\beta_m + \gamma_m^s \rho) w_m - \delta_m((1 - a_f)(w_f e_f))}{(\beta_m + \gamma_m^s \rho + \delta_m) w_m}$$

$$p_m x_m^s = \frac{\beta_m}{\beta_m + \gamma_m^s \rho + \delta_m} (w_m + (1 - a_f^s)(w_f e_f^s))$$

When it is a daughter, the wife the child's entire expense. Then in equilibrium,

$$y_f^d = \frac{\gamma_f \rho}{\beta_f + \gamma_f \rho} a_f (w_f e_f)$$

$$\frac{\beta_f + \gamma_f \rho}{a_f} = \frac{\alpha_f}{2a - a_f} \rightarrow a_f^d = \frac{\beta_f + \gamma_f \rho}{\alpha_f + \beta_f + \gamma_f \rho} 2a$$

Similarly,

$$e_f^d = \frac{\beta_f + \gamma_f \rho}{\beta_f + \gamma_f(\lambda + \rho) + \delta_f}, \quad h_f^d = \frac{\gamma_f \lambda}{\beta_f + \gamma_f(\lambda + \rho) + \delta_f}$$

The husband:

$$e_m^d = \frac{\beta_m w_m - \delta_m((1 - a_f)(w_f e_f))}{(\beta_m + \delta_m) w_m}$$

$$p_m x_m^d = \frac{\beta_m}{\beta_m + \delta_m} (w_m + (1 - a_f^d)(w_f e_f^d))$$

Her consumption change is unclear.

$$p_f x_f^s = \frac{\beta_f}{\alpha_f + \beta_f} \frac{\beta_f}{\beta_f + \gamma_f \lambda + \delta_f} w_f$$

$$p_f x_f^d = \frac{\beta_f}{\beta_f + \gamma_f \rho} \frac{\beta_f + \gamma_f \rho}{\alpha_f + \beta_f + \gamma_f \rho} \frac{\beta_f + \gamma_f \rho}{\beta_f + \gamma_f(\lambda + \rho) + \delta_f} w_f$$

Lastly, it is easy to check that the wife's overall utility is lower when it is a girl. When it is a girl, the wife's choices are $(x_f^d, e_f^d, h_f^d, y_f^d, a_f^d)$. When it is a son, using the same set of choices plus $y_m^d > 0$ must give the wife a strictly higher utility. ■

B.5 A collective model

In a collective model, a social planner, representing the wife and the husband, maximizes the weighted total utility: $U = a_f u_f + (1 - a_f) u_m$. Recall the wife and the husband's

utilities are ($y = y_f + y_m$):

$$u_f = \beta_f \ln x_f + \gamma_f (\lambda \ln h_f + \rho \ln y) + \delta_f \ln(1 - e_f - h_f)$$

$$u_m = \beta_m \ln x_m + \gamma_m^{s,d} (\lambda \ln h_f + \rho \ln y) + \delta_m \ln(1 - e_m)$$

In India, husbands are in dominant positions, so we maintain the assumption that $u_m > u_f$ in equilibrium. Thus, the social planner has incentives to give the husband a higher weight once a_f is endogenous. In particular, continue using $\underline{a}$ as the baseline autonomy determined by spouses' outside options, and the social planner can costly move autonomy away from its baseline value. Let Δa_f be the distortion towards the husbands from baseline value, s.t. $a_f = \underline{a} - \Delta a_f$. The cost to do so is $\alpha \ln(1 - \Delta a_f / \underline{a})$, such that there is no cost if there is no distortion. Overall, the social planner's utility is

$$U = (\underline{a} - \Delta a_f) u_f + (1 - \underline{a} + \Delta a_f) u_m + \alpha \ln(1 - \Delta a_f / \underline{a})$$

Suppose when it is a son, the social planner's optimal choice is Δa_f^s , u_f^s and u_m^s . Note that by assumption, $u_f^s < u_m^s$, and thus the optimal $\Delta a_f^s > 0$. When it is a daughter, the husband cares less of the household public good (the well-being of the child), $\gamma_m^d < \gamma_m^s$. The direct effect of a daughter is decreasing the husband's utility while keeping the wife's utility constant. To fix the idea, suppose the decrease of the husband's preference $\gamma_m^s - \gamma_m^d$ is sufficiently high, such that the equilibrium adjustment doesn't alter the direct effect. Then the gap of spouses' utilities is smaller when it is a daughter, $u_m^d - u_f^d < u_m^s - u_f^s$. The FOC suggests that

$$u_m - u_f = \frac{\alpha}{\underline{a} - \Delta a_f}$$

So $\Delta a_f^d < \Delta a_f^s$. The wife gets a higher autonomy with a daughter, because of a lower distortion.

B.6 General utilities

Finally, we show that our main results do not depend on the particular log linear form of utilities. Instead, we consider the general utilities for the husband and wife:

$$u_m(x_m, y, l_m), \text{ and } u_f(x_f, h_f, y, l_f) - c(a_f),$$

in which $y = y_f + y_m$, $p_m x_m = w_m e_m + (1 - a_f) w_f e_f - y_m$, $l_m = 1 - e_m$, $p_f x_f = a_f w_f e_f - y_f$, and $l_f = 1 - e_f - l_f$. We assume these utility functions are increasing, differentiable

and concave, and the cost of autonomy is increasing, differentiable and convex. Moreover, we assume the husband's wage is higher, $w_m > w_f$, to make sure the change in the husband's behavior will not be fully offsetted by the wife's response.

Proposition B.10. *Suppose the Nash equilibrium is interior and unique, and the cost $c(a_f)$ is sufficiently high. If the husband's marginal preference for the public good decreases sufficiently for a daughter than a son $\left(\frac{\partial u_m^s}{\partial y}(\cdot) > \frac{\partial u_m^d}{\partial y}(\cdot)\right)$, then when the child is a daughter (relative to son):*

- *The wife works more outside (e_f), less at home (h_f), and gets a higher autonomy (a_f).*
- *The husband works less outside (e_m).*
- *The monetary investment to the household public good decreases ($y_f + y_m$).*

Proof of Proposition B.10: Normalize $p_m = p_f = 1$ since they do not impact the results. Consider the FOCs for the husband:

$$[e_m, y_m] : \frac{\partial u_m}{\partial x_m} w_m = \frac{\partial u_m}{\partial l_m} \text{ and } \frac{\partial u_m}{\partial x_m} = \frac{\partial u_m}{\partial y}.$$

The FOCs for the wife are:

$$[e_f, h_f, y_f, a_f] : \frac{\partial u_f}{\partial x_f} a_f w_f = \frac{\partial u_f}{\partial l_f}, \frac{\partial u_f}{\partial h_f} = \frac{\partial u_f}{\partial l_f}, \frac{\partial u_f}{\partial x_f} = \frac{\partial u_f}{\partial y}, \text{ and } \frac{\partial u_f}{\partial x_f} w_f e_f = \frac{\partial c}{\partial a_f}.$$

First, we claim that compared to a son, when it is a girl, in equilibrium, $\frac{\partial u_m}{\partial y}$ must decrease. Note that the exogenous change is for any fixed values of all parameters, $\frac{\partial u_m}{\partial y}(\cdot)$ decreases, but between these two equilibria, the spouses could respond by decreasing y which increases the marginal preference. Since we assume the exogenous decrease is sufficiently high such that to bring the marginal preferences back to the level with a son, y has to decrease a lot. Based on the wife's FOC, it implies y_f , x_f , l_f and h_f all have to decrease a lot. But then, e_f must increase and $a_f \geq \underline{a} > 0$. Then it violates the wife's budget balance, since her earning does not decrease much, but she spending decreases a lot. The endogenous adjustment does not fully undo the decrease of $\frac{\partial u_m}{\partial y}$. This implies, by the husband's FOCs, e_m decreases.

Second, we claim y decreases when it is a girl. This is because the decrease in the husband's preference is sufficiently high, such that y_m decreases a lot. By our assumption, $w_m > w_f$, and the wife only controls a part of her earnings, so the wife does not have the ability to fully make up for the decrease in y_m .

Lastly, we show the first bullet point on the change of the wife's behavior. By our assumption that $c(a_f)$ is high, in both equilibria, $a_f - \underline{a}$ is not too large. Then by $\frac{\partial u_f}{\partial y} a_f w_f =$

$\frac{\partial u_f}{\partial l_f}$, y decreases a lot while a_f is not far from $\underline{a}$, so l_f must decrease. It implies h_f decreases, and then e_f must increase. Lastly, by $\frac{\partial u_f}{\partial y} w_f e_f = \frac{\partial c}{\partial a_f}$, y decreases and e_f increases, so a_f increases. ■

C Other effects of daughters on their mothers

C.1 Daughters affect subsequent fertility and their mothers' health

Many effects of a daughter would likely hurt her mother's outside option, such as the health effects discussed in section 4.2.2. In any case, given the generally consistent evidence that daughters tend hurt their mother's health, it seems unlikely that health effects could actually improve a mother's autonomy. The higher subsequent fertility a daughter prompts could also affect a mother's autonomy directly by changing her outside option, but if anything, more children would likely decrease a mother's autonomy by making it harder for her to survive on her own, given biological reasons and social norms that children would remain with the mother upon dissolution of a marriage. Overall, we argue that the tendency for daughters to have more siblings cannot explain the increases in autonomy we see.

C.2 Daughters affect their mothers' preferences or beliefs

A final potential alternative explanation is that a daughter directly affects her mother's preferences or beliefs. For instance, a daughter may learn about gender equality in school and convince her mother to stand up for herself. Or as the mother observes her daughter's opportunity set (which would often be larger than her prior beliefs if gender equality is increasing over time), the mother updates her beliefs on possible outcomes for women. There is indeed some evidence that daughters decrease the probability that their mothers report that domestic violence is ever acceptable. Panel A of table A8 shows that in Bangladesh, a daughter is associated with a 0.70 percentage point decrease in the probability her mother reports that domestic violence is ever acceptable in the sample of all births; a firstborn daughter is associated with a 1.1 percentage point decrease ($P = 0.106$). In India each daughter is associated with a 0.44 percentage point decrease in the probability the mother believes domestic violence is ever acceptable; a firstborn daughter decreases the probability by 0.57 percentage points ($P = 0.188$).

A natural question is whether these suggestive changes in mother's perceptions of domestic violence lead to actual changes in domestic violence. The India survey asked

women about actual incidence of domestic violence, and we report the effects of daughters on domestic violence in Panel B of table [A8](#). While there is evidence in the sample of all children that the number of daughters is associated with lower rates of (moderate) domestic violence, this effect goes away in the sample of only firstborn children, so we do not interpret it as causal effect of daughters. Countervailing effects could explain the zero effect in the sample of firstborn children. For instance, mothers' increased perception that domestic violence is never justified could be translated into less risk of domestic violence, but this effect could be counteracted by the possibility that the mother seeking more autonomy leads to frictions that result in violence, as in the models of [Eswaran and Malhotra \(2011\)](#) and [Anderson and Genicot \(2015\)](#). As we discuss in section [2.1](#), this violence could be one component of the cost of seeking autonomy in our model.

The fact that the mother's apparent decreased belief in the acceptability of domestic violence does not translate into lower domestic violence (in India) suggests that changes in preferences from daughters do not automatically translate into better treatment, and thus that changes in preferences or beliefs prompted by daughters are unlikely to singlehandedly explain increases in autonomy. Additionally, increases in beliefs in gender equality cannot explain the marginal decreases in consumption without incorporating other effects of daughters such as the differential preferences for spending on daughters posited by our model. In sum, while we do find some evidence that daughters affect their mothers' perceptions of gender relations, we nonetheless think that it is unlikely that this model can explain the whole set of results we find.